WORKING TITLE
Linguistic tells and business themes in the trademark record

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Abstract

All 13.99 million USPTO trademark case files are re-parsed into event-dated prosecution histories, and every filing is scored on how surprising its goods/services language is against what its industry filed in the five years before and the five years after. The average of the two surprises is the filing's atypicality; their difference is its lead, positive for wording the industry then moved toward. Among 3.10 million registrations, the most leading fifth are cancelled at the five-year proof of continued use 1.4 points more often than the most lagging fifth, within Nice class and registration year with drafting and counsel held ($t = 8.3$). The penalty holds under fifty global themes, five hundred, and fifty themes fitted per class. It attaches to language new to the whole record: a theme established elsewhere and extended into a new Nice class carries no penalty — those extensions survive the gate slightly better than the class's own established themes, and they are most of what a global model scores as atypical. The penalty was two to three times larger for marks filed in the late 1990s, reversed for 2000–2004 filings, and returned from 2008, with the swing inside four technology classes. Among one million owners debuting 2009–2018, the debut's language does not predict a priced round; SEC reporting and listing fall in lead. The firms holding most linked revenue described their first product in language their industry already used. Applicants who refile an abandoned mark rewrite toward the class's typical description, so every registered description is in part a conformed one. Code, panels, and the corpus build are public at `https://github.com/ericsilver/tm-vocabulary`.

1 Introduction

The pioneers are the ones with arrows in their backs. The claim has been tested on named markets and naming choices (Golder & Tellis, 1993; Semadeni & Anderson, 2010), but not on a record where being early can be read directly from what every entrant wrote, because that record did not exist in usable form. It does now. Every US trademark application since 1884 is public, with its goods/services description, its owner, its counsel of record, and a dated prosecution history that records whether the mark registered, whether its owner proved five years later that the product was still in commerce, and whether it was renewed at ten. This paper re-parses all 13.99 million case files into that dated form and asks of each filing two questions about its language: how unusual was this description for its industry, and was it written in words the industry was moving toward or words it was leaving behind. The first is the filing's *atypicality*, the second its *lead*. The paper then follows the filing, and the firm behind it, through the gates the record provides.

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The sections are written to be read on their own: a reader can take the definitions of Section 2 and then the one section their question lives in. Four kinds of reader are anticipated. A reader interested in trademark prosecution — how applications are drafted, examined, and rewritten — needs Section 3. A reader interested in whether new industries, fads, and technology waves produce firms that fail faster needs Section 5. A reader interested in whether the language of a first filing predicts commercial success needs Sections 4 and 6. A reader interested in the measurement itself needs Section 2 and the robustness of Section 8.

Section 2 builds the measure. A description is reduced to the themes it draws on, fitted once over the whole corpus by a topic model and held fixed; its theme proportions are compared, by Kullback–Leibler divergence, to the average proportions of its own Nice class — the international classification of goods and services under which every application is filed, forty-five classes, a filing claiming one or several — in the five years before its filing date and the five years after. The average of the two divergences is atypicality; their difference is lead, written L. The section states why words are not scored directly (word-level divergence rises mechanically with description length), why surprise is measured within a class and never across classes, what the measure does not see (how a filing combines its themes), and the record's gates and counting conventions, stated once for all the sections that follow.

Section 3 follows every filing to registration. Registration rises with atypicality and turns down only in the most atypical decile, is nearly flat in lead — a quantity no examiner could observe, since it is computed from filings not yet made — and differs sharply by who drafts: 62% of counsel-represented applications register against 46% of self-filed ones. Applicants who refile an abandoned application rewrite toward the class's typical description, from both tails, whoever drafts the second attempt, so every registered description is in part a conformed one.

Section 4 is the paper's central result. Among 3.10 million registrations, marks whose language most led their industry were cancelled at the five-year proof of continued use 1.4 points more often than marks whose language most lagged it, within Nice class and registration year, with drafting and counsel held — and 2.3 points raw, in 33 of 44 classes, robust to the timing of failure, to who files, to atypicality strata, and to three separate partitions of the vocabulary. Atypical marks survive the proof more often, but that protection belongs mostly to filings extending a theme established elsewhere into a new class; net of atypicality, even those fail more. The risk does not attenuate at the year-ten renewal.

Section 5 locates the penalty in time and industry. It was two to three times larger for marks filed in the late 1990s, reversed for marks filed 2000–2004, and returned from 2008, with the whole swing inside four technology classes and, within them, in which themes each era's leading filings were drawn from. Web-bearing filings fail hardest in each class's own first internet years and converge toward their class within a decade. Surging themes cost their joiners little, and arriving early within a wave costs nothing; the one corpus-scale wave, online software in 1999, is the exception.

Section 6 follows the owner rather than the mark. On one million owners making their first registered filing in 2009–2018, the debut's language does not predict a priced private round; SEC reporting and listing fall in lead, and the listings that arrive without an observed round belong disproportionately to lagging debuts. The firms that came to hold most of the linked value (ten owners hold 18.5% of \$19.2 trillion of peak revenue) described their first product in language their industry already used. Neither axis is a proxy for patenting. Section 7 reads the stages together on one harmonized sample; Section 8 reports what changes under other choices of resolution, weighting, window, and period — one choice, the decayed reference window, shifts the headline estimate, by about a seventh.

Being early with the language of a product costs at the five-year proof of continued use, in three of every four product classes and under every partition of the vocabulary, and it costs most in the industries and the years in which the language was moving fastest. Being unusual is rewarded at the same proof, mostly where a theme is imported from elsewhere, and it is the examiner, not the market, that rewards middling conformity at registration. The firms that thrived were not the pioneers of language. They were written in the words their industries already used, and they are few.

2 Context, approach, and vocabulary

2.1 The record and the question

A US trademark application carries a description of the goods or services the mark will cover. The description is written to secure scope of protection: wide enough to cover the planned use, narrow enough to be granted. It is therefore a forced, dated disclosure of what a firm expected to sell. The USPTO publishes every application since 1884 with its prosecution history, and this paper re-parses all 13.99 million case files into dated events, so that each application can be followed from filing to registration, to the sworn proof of continued use required between the fifth and sixth year, and to cancellation or renewal.

The question is whether the language of the description, read against the language of the rest of the industry at the time, predicts what happened to the mark and to the firm behind it. Two properties of the language are measured. *Atypicality* is how unusual the description is for its industry. *Lead* is whether its language is ahead of or behind where the industry's language was moving. Sections 3–6 relate the two to reaching registration, to the five-year proof of continued use, and to venture funding and public listing. Section 8 reports what changes under other choices.

2.2 Two readers, one living backwards

Imagine two readers of a filing's goods/services description, each having seen a disjoint half of its industry. One has read only what the industry filed in the five years before it. The other, living backwards as Merlin was said to, has read only the five years after. Each is asked how surprising the wording is. The measurement is how much they disagree: a description that startles the first reader and not the second arrived before its industry's language moved that way.

Both windows are anchored on the filing's own date, spanning the 1,826 days before it and the 1,826 days after; both exclude the filing date itself, so a filing never appears in the reference of another filing made the same day, and no two filings made on different days share a reference. Inside a window every day counts the same. The measure asks whether wording is unusual against a period, not whether it is ahead of a trend inside that period; it has no momentum channel.

2.3 What is measured

The quantity is information content in Shannon's sense. Something that happens with probability p carries $-\log p$ of information when it happens: a common thing carries little, a rare thing a lot. Here the things are the themes a description draws on, and the probabilities are how often the filing's own industry drew on each of them in the reference window. A filing's score is the average information content of its themes under its industry's frequencies, weighted by how heavily it uses each, less the same average taken under the filing's own frequencies.

The apparatus is that of Murdock et al. (2017), developed on Darwin's reading notebooks, and Barron et al. (2018), who extended it to French Revolution parliamentary debates. They score a document against those before it and again against those after, calling the two quantities *novelty* and *transience* and their signed difference *resonance*. Their object was the life of ideas: how often an idea appeared, and how ideas stood in relation to one another, across one reading life or one assembly's debate. The object here is the composition of commercial offerings, which components a product or service is described as having, and how that composition shifts across an industry over time. The arithmetic carries over unchanged. The interpretation does not: their document has one author and the reference stream is that author's own reading, so a high value says something about a mind; here a filing is scored against other firms' filings, and its author is usually counsel drafting for scope of protection.

Themes, not topics. The clusters the model fits are called *themes* throughout. In the source literature and in the name of the method, latent Dirichlet allocation, the same objects are called topics, and a document's topic is what it is about. A goods/services description is not about anything in that sense; it lists components. A cluster of terms that co-occur across descriptions (*video, music, games, electronic, audio*) is a component that many offerings share, not a subject. Theme is the word for that. Topic is used here only in the name of the model.

2.4 From a description to two numbers

Counting words. A description is reduced to a bag of words: the text is lowercased, split into tokens of three or more letters, and counted, with adjacent pairs counted as tokens in their own right so that *computer software* registers alongside *computer* and *software*. Order is discarded. Take the filing for Amazon.com, lodged in the advertising and retail class on 18 April 1997:

> computerized on line search and ordering service featuring the wholesale and retail distribution of books, music, motion pictures, multimedia products and computer software in the form of printed books, audiocassettes, videocassettes, compact disks, floppy disks, CD ROMs, and direct digital transmission

Of its 45 distinct terms, 36 are in the theme model's vocabulary, among them *computerized*, *search*, *ordering*, *ordering service*, *retail*, *distribution*, *printed books* and *digital transmission*.

Grouping words into themes. Each filing is then re-described in a fixed, much smaller set of coordinates. Latent Dirichlet allocation is run once over a stratified sample of the whole corpus, all 45 classes together: it learns 50 clusters of words that tend to occur together and, for any filing, the proportions in which its words draw on them. Nothing supervises this; the clusters come from co-occurrence alone. This build — fifty global themes, per-filing five-year windows, flat weighting — is the production scoring, used for every estimate unless the text states another. The themes are shared across classes; what is specific to a class is the reference, the average theme proportions of that class's own filings in the window. The Amazon filing draws on six themes at more than a twentieth of its weight: 0.41 on media and entertainment goods (*video, computer, electronic, music, games, audio*), 0.15 on computer and information services (*services, computer, software, information, data*), 0.13 on retail store services (*store, retail, store services, featuring*), 0.10 on printed matter (*paper, books, printed, cards, stationery*), 0.09 on downloadable and online software, and 0.07 on marketing of consumer goods. The online appendix lists every theme with its leading words and its weight in each class.[1]

Fifty is coarse, roughly one theme per industry. Section 4.4 refits the model at 500 themes and separately at 50 themes within each class and repeats the five-year-proof analysis under each;

[1] `https://aporia.institute/tm-vocabulary/online-appendix/themes_T50.html`

Section 8 reports the resolution sweep and a second fit at the same resolution with a different seed. Fifty is kept elsewhere because it is the coarsest resolution at which every result holds, and because its software-class lead contrast sits at the low end of the resolutions tried (only the 100-theme refit returns less).

Comparing against the two references. Write P for the filing's distribution over themes, and Q_{past} and Q_{future} for that distribution averaged over every same-class filing in the two windows, $[d_i - W, d_i)$ and $(d_i, d_i + W]$, with $W = 1{,}826$ days. How far P sits from a reference Q is the Kullback–Leibler divergence,

$$\mathrm{KL}(P \,\|\, Q) \;=\; \sum_k P_k \log \frac{P_k}{Q_k},$$

a sum over the 50 themes, zero when the two distributions agree and growing as the filing puts weight where the industry puts little. It is measured in nats, the natural-log unit. Write

$$K^- = \mathrm{KL}(P \,\|\, Q_{\text{past}}), \qquad K^+ = \mathrm{KL}(P \,\|\, Q_{\text{future}}),$$

so K^- is what the first reader feels and K^+ the second: the filing's past-facing and future-facing surprise.

Taking the difference. Amazon's filing is scored against 41,166 earlier filings and 146,532 later ones, giving $K^- = 1.286$ and $K^+ = 1.163$. The two axes used throughout are their average and their signed difference:

$$\underbrace{A \;=\; \tfrac{1}{2}\left(K^- + K^+\right)}_{\text{atypicality}}, \qquad \underbrace{L \;=\; K^- - K^+}_{\text{lead}}.$$

For Amazon, $A = 1.225$ and $L = +0.123$: against its own class, its atypicality is at the 47th percentile and its lead at the 94th. The industry moved toward this description in the five years after it was filed and had not been writing that way in the five years before. Table 1 maps the terms; the signed difference is written L throughout and its magnitude $|L|$.

What "early" means is visible in the themes themselves. Of the six Amazon drew on, the share of class-035 filings drawing on each, in the five years before its filing and the five after:

Theme	Amazon's weight	Before (%)	After (%)
Retail store services	0.13	42.6	50.3
Computer and information services	0.15	58.9	59.9
Media and entertainment goods	0.41	9.9	11.3
Downloadable and online software	0.09	2.3	4.1
Printed matter	0.10	8.3	7.8
Marketing of consumer goods	0.07	10.5	9.3

A theme counts as drawn on when it carries at least a tenth of a filing's weight. The three themes that carry most of Amazon's description, retail, media goods, and the still-small online-software theme, all became more common in the class over the following five years, the last nearly doubling; the two minor themes it also touched became slightly less so. The filing is closer to the class's future composition than to its past one.

K^- and K^+ correlate at 0.988 on the software class, so entering both is close to entering one variable twice; rotating them into an average and a difference isolates the large common component from the small residual that carries the timing, and leaves the two regressors nearly uncorrelated (+0.036 against each other, where K^- alone would give +0.114). The cost is that L is a small residual of two large quantities, about a sixth of either level's spread, which Section 8 quantifies.

Table 1: The quantities, their sources, and the terms used here. *Lagging* and *leading* describe the two signs of L; *atypicality* is unsigned and says nothing about direction.

Quantity	Definition	Murdock et al.	Term used here
Surprise against the past	K^-	novelty	past-facing surprise
Surprise against the future	K^+	transience	future-facing surprise
Their average	$A = \frac{1}{2}(K^- + K^+)$	—	**atypicality**
Their signed difference	$L = K^- - K^+$	resonance	**lead** (leading / lagging)
Its magnitude	$\lvert L\rvert$	—	lead magnitude

Why the words are not scored directly. The themes are an extra step, and the words themselves could be compared to the reference: build the filing's distribution over the class vocabulary from the words it uses, do the same for every filing in the window, and take the divergence. The paper reports that scoring as a cross-check. It cannot be the primary one, because a divergence measured on words mostly counts the length of the filing. The divergence is an average surprise less a spread: for any two distributions,

$$\mathrm{KL}(P \,\|\, Q) \;=\; \underbrace{-\textstyle\sum_k P_k \log Q_k}_{\text{how unexpected these words are}} \;-\; \underbrace{\left(-\textstyle\sum_k P_k \log P_k\right)}_{\text{how spread out the filing is}} \,,$$

and the spread of a filing over words is bounded by how many distinct words it has: a description using k terms cannot spread over more than k of them, so its spread cannot exceed $\log k$, while the reference is spread over about 900 terms. A five-word description is charged the whole difference in width, roughly five nats, before any of its content is considered. In 3,000 software-class filings of at least 150 words, scored against one fixed reference and rescored after truncation to a random handful of words, across a fiftyfold cut in length the surprise term moves by 0.016 nats and the spread term by 2.9. Synthetic filings whose words are drawn at random from the reference itself, maximally ordinary by construction, score 5.72 at three tokens and 1.65 at four hundred, tracking log(distinct terms) to within a fiftieth of a nat, and the average real filing scores marginally below that null at its own length. Term-scored atypicality correlates with the log of a filing's distinct term count at -0.68 in the software class.

Themes break the coupling because the 50 coordinates are always occupied. Under term scoring the effective number of coordinates a filing spreads over rises seventeenfold from the shortest descriptions to the longest; under themes it varies by less than a third, and not monotonically: a three-word fragment gives the model little evidence, so its inferred proportions stay close to the prior and spread across many themes. The spread term no longer tracks length, and the score is dominated by the surprise term. Theme-scored atypicality correlates with the log term count at -0.22 in the software class and -0.10 in advertising and retail, against -0.68 and -0.62 under terms.

What the themes see, and what they do not. The score compares a filing's theme proportions to the average proportions of its industry, so it reads which themes a filing draws on and how heavily. It does not, in any material way, read how a filing combines them. Taking each filing's two leading themes and asking how often that pair occurs together relative to the two themes' separate frequencies, on a 150,000-filing software-class sample: how rare the themes are individually explains 70% of the variance in atypicality, and the unusualness of the pairing adds under two points on top of it. A filing assembled from ordinary parts in a genuinely new configuration is therefore close to invisible to this measure.

Combinations can be scored in their own right, at two levels, and the two levels behave differently. Both are computed on the software class, and both are reported net of the filing's own lead and atypicality, so each is the increment over what the construct already measures.

At the theme level, take every pair of themes a filing draws on (at a tenth of its weight or more) and measure how much more or less often each pair occurs together than the two themes' separate frequencies would predict, in the class's filings of the five years before and the five after.[2] The negative log of that ratio is the pairing's information content; averaging over a filing's pairs, weighted by their masses, and differencing the two windows gives a *pair lead*, positive when the class moved toward the filing's pairings. Filings in the top fifth of pair lead fail the five-year proof of continued use 2.3 points more often than those in the bottom fifth ($t = 8.4$, unchanged by the controls), and pair lead correlates with lead itself at only -0.18. Being early with an arrangement of themes is penalised in its own right, at a rate comparable to being early with the themes. The rarity of a filing's pairings, by contrast, carries nothing once its atypicality is held (-0.3 points, $t = -1.1$).

At the word level the sign reverses. For each filing, count the share of its adjacent word pairs that are new to the class — the pair appearing fewer than five times in the class's prior five years — while both words are established there, each appearing at least fifty times; the share is taken over such eligible pairs only. This is recombination in the sense of the innovation literature, novelty assembled from familiar components (Fleming, 2001; Uzzi et al., 2013): familiar parts, new adjacency. Filings in the top fifth of that share fail the five-year proof 6.1 points *less* often than those in the bottom fifth, and 2.8 points less with lead and atypicality held ($t = -10.4$). Word recombination and theme-pair lead are uncorrelated ($r = -0.01$): they are different facts about a filing. What the record shows, then, is not that combination is invisible and neutral. Arranging themes the industry had not yet arranged is punished like arriving early; coining phrases from the industry's own words is rewarded. The construct the paper uses sees neither, and the two run in opposite directions, so its blindness to combination is a bounded omission rather than a hidden bias in one direction. The sections that follow use lead and atypicality of the parts; these two measures bound what they leave out.

2.5 Within class, not across it

Goods/services descriptions are drafted to secure scope of protection, not to describe a product distinctively, and counsel reuse language from the USPTO's own Acceptable Identification of Goods and Services Manual, from house precedent, and from competitors' successful filings. Identical wording across unrelated firms is therefore common, and it carries through to the scores: 17% of scored registrations issued 2002–2018 share a value with another filing in the same class and registration year. Within a class, the measure picks up departure from a shared drafting convention. Across classes it does not: a class where the Manual supplies dense standard language shows a compressed distribution of atypicality, and one where firms describe novel services in prose a wide one, for reasons unrelated to how innovative either industry is. Every estimate is therefore taken within class and year, and magnitudes compare across classes only as signs and orderings.

The vocabulary is positional rather than causal. A leading filing was ahead of its industry's language, which implies a trajectory without asserting the filing caused it. Atypicality is textual, and is not trademark law's distinctiveness doctrine, which governs registrability along the generic-to-fanciful spectrum; the two attach to different objects, one to the mark and one to the description of the goods it covers.

[2]The pair windows are calendar-year buckets rather than per-filing days, and a window is scored only when it holds at least 2,000 filings.

2.6 The gates, the samples, and the conventions

The outcome sections that follow are written to be read independently, so the record's steps, the outcome names, and the counting conventions are set out once, here.

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a sworn statement of use is filed — the proof that the mark has entered commerce — the mark registers. Registration starts a maintenance clock: in years five to six the owner must file a Section 8 declaration proving continued use in commerce, with a six-month grace period — the *five-year proof of continued use*, the paper's central outcome; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a proof is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific proof by its date rather than inferred from a status snapshot.[3]

Table 2: The trademark funnel in this corpus. Registration counts are unique serials; five-year-proof outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Source: the re-parsed USPTO application backfile, scored as in this section.

Stage	n	Share
Debut filings scored, 1995–2018 (owners' first filings)	3,589,068	
reached registration	2,066,600	57.6%
All unregistered class-records filed 1995–2018	3,798,318	
never filed statement of use		42.6%
stopped responding in prosecution		45.3%
removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the five-year proof of continued use (event-dated)		52.3%
Passed the five-year proof, cohorts 2002–2013, status known	1,129,724	
failed at the year-ten renewal		39.1%

Three named outcomes recur (Table 2). *Failing to register*: the application never completed, which happens by the applicant's own inaction in roughly nine cases of ten (Section 3). *Failing the five-year proof of continued use*: a registration cancelled for non-use at registration age 4.0 to 8.5 years, which means the product the mark named left commerce within six years (Sections 4 and 5). *Reaching the SEC record*: the owner appears in dated SEC events — a priced Form D round, financial reporting, a listing (Section 6).

Outcomes are rates, so every estimate is a shift in a probability and is stated in percentage points against the base rate it shifts: +1.4pp on a 52.3% base means 53.7% against 52.3%. The basic comparison is the outcome rate of the fifth of filings with the highest lead minus that of the fifth with the lowest, with the fifths cut inside a single Nice class and year so that no difference between industries or between years enters; regression tables add drafting and filer controls and cluster errors on the owner. Where a continuous version is given, it is the change in the outcome for a one-standard-deviation change in the axis, in percentage points.

[3]Cross-checked against the USPTO status-code dictionary: 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

Counting differs by table. A filing may claim several Nice classes at once, so a *class-record* counts it once per class while a *registration* counts it once by serial number; a *debut filing* is an owner's first, one per owner. A *cohort* is the set of registrations issued in a calendar year; where a split is instead by filing year, the text says so.

Two windows bound the samples, each set by an institutional fact. Scoring runs 1995–2019: a full five-year past reference does not exist before 1990 and a full forward reference does not exist after 2020, and the 1995 cut is about two years more conservative than the measured burn-in requires (Appendix A). Five-year-proof cohorts run 2002–2018: cancellations post at about registration age six and a half, the record ends in April 2026, so 2018 is the last cohort whose window has essentially elapsed, and 2002 is the conventional lower cut for this corpus — a bound that matters, because carried back to the first scoreable cohorts the penalty was larger than it has ever been since (Section 5.1).

2.7 What the construct is not called

Novelty and *transience*, in the sense of Murdock et al. (2017) and Barron et al. (2018), are the two KL levels themselves: novelty is divergence from the past, transience divergence from the future. Atypicality as used here is their average, and so is neither of them alone; *novelty* in particular is declined because it names only the backward-looking half of a quantity that faces both ways. *Resonance* is their signed difference, and is arithmetically what this paper calls lead. The word is avoided because it names a mechanism — the field resonating with an author's contribution — that this design cannot observe. A filing can sit where its class was heading because it was imitated, because it and the class responded to the same outside event, or by coincidence; the measure does not separate these, and *resonance* would assert the first.

Innovation is a property of the product, not of the sentence describing it: two firms selling the same thing can draft very differently, and a genuinely new product can be described in boilerplate (Appendix E). The measure sees prose.

Radicalness (Dewar & Dutton, 1986; Henderson & Clark, 1990; Tushman & Anderson, 1986) is the closest neighbour, since KL is a departure measure rather than a newness count and the Amazon exhibit is a large departure built from unremarkable components. It is declined because it claims that existing competences are devalued, and nothing here licenses the step from unusual language to devalued competence.

Distinctiveness has two prior occupants. In trademark law it is the registrability doctrine, proven in part by the continued use this paper uses as an outcome. In strategic management, optimal distinctiveness (Zhao et al., 2017) predicts an inverted-U in conformity — a shape that appears here at registration but not at listing, so the term would imply a theory is being tested when it is not.

What *atypicality* says is narrower than any of them: this description is unusual for its class and year. Whether the thing described was new, radical, or good is not measured.

3 Filing, drafting, and reaching registration

This section is about the first step only: whether an application reaches registration, on every class-record filed 1995–2018. Later steps — whether the product stayed in commerce, and what became of the firm — are Sections 4–6 and do not enter here.

3.1 Registration rises with atypicality and is nearly flat in lead

Figure 1 gives the registration rate by decile of each axis, with deciles cut within Nice class and filing year so that no difference between industries or between years enters. On atypicality the rate rises from 56.0% in the least atypical decile to 59.1% in the ninth and turns down to 58.3% in the most atypical — a three-point climb and a one-point turn at the top. The turn is larger in some industries than the pooled line shows: software and electronics rises through the whole range (51.8% to 60.6%), clothing dips in the middle deciles, and both end higher than they start. On lead the range is 0.9 points with no consistent direction: filings written in language their class was moving toward register at nearly the same rate as filings written in language it was leaving.

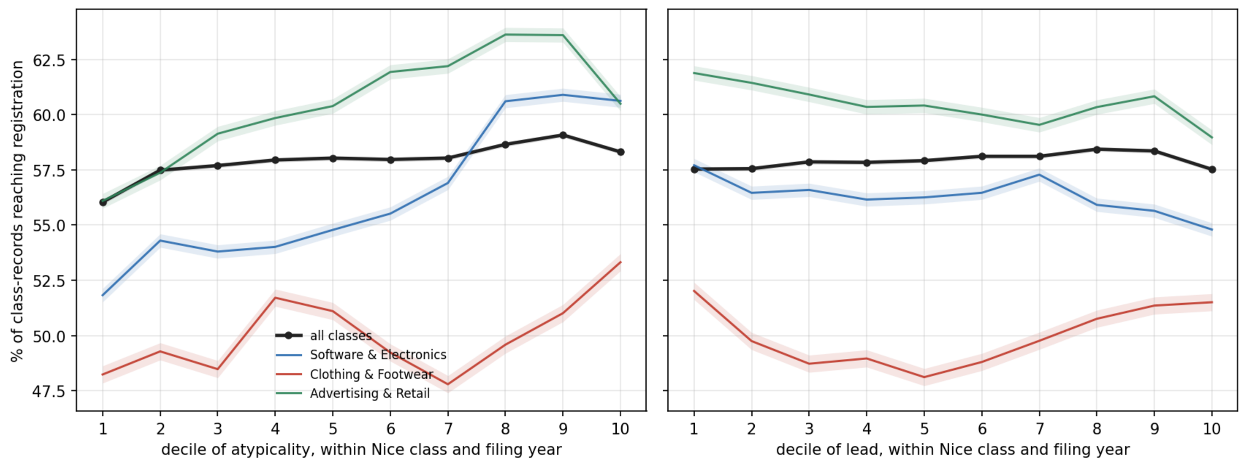

Figure 1: Share of class-records reaching registration, filings 1995–2018, by decile of atypicality (left) and of lead (right), deciles cut within Nice class and filing year; $n = 9{,}024{,}532$ scored class-records. The black line pools all 45 classes; the colored lines are three named classes. The grey band is a 95% binomial interval on the pooled line.

The flatness of lead at this step has a mechanical reading worth stating: lead is computed from the five years of filings *after* the filing date, which had not happened when the examiner acted. No participant in prosecution could observe the quantity. Whatever small association appears must run through correlates that were visible at the time — the words themselves, the industry, the drafter. Atypicality's past-facing half, by contrast, is exactly what an examiner steeped in the class's usual language experiences, and it is the axis registration responds to.

Figure 2 shows the same outcome on the two surprise axes directly, for the two largest classes: each hexagonal cell is colored by the share of its filings that registered. The color gradient runs along the diagonal — from the low-surprise corner, where registration is lowest, up through the middle of the range — and barely across it. Distance from the diagonal is lead; position along it is atypicality. The plane says graphically what the deciles say numerically: the examination step reads how far a filing sits from its class's usual language, not which direction it faces in time. The two marked filings are discussed as worked examples in Section 2 and Appendix E.

Two cautions on reading the level. First, registration is mostly a measure of the applicant's own follow-through: of the applications that do not register, roughly nine in ten die by inaction — the applicant never files the statement of use, or stops responding during prosecution — and only 4.3% are removed adversarially (Table 2). Second, cut on the pooled distribution rather than within class and year, and restricted to debut filings, the atypicality curve is an inverse-U with a deeper conventional tail (46.6% at the bottom, about 58% in the middle, 53.4% at the top); that version

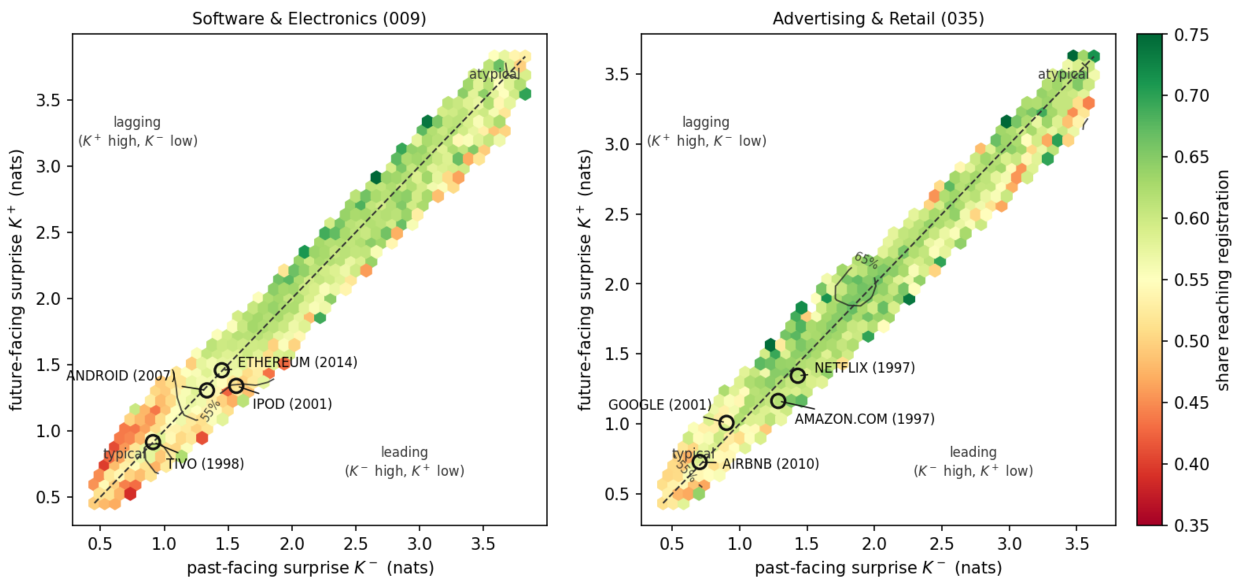

Figure 2: Share reaching registration across the surprise plane, filings 1995–2018 in the two largest classes. Each hexagon holds at least 50 class-records and is colored by the share of them that registered; the dashed line is $K^- = K^+$, on which lead is zero. Cells below the line are leading, cells above lagging.

carries class and cohort composition, and Appendix B reports it with the reasons familiarity alone cannot produce the shape. The within-cell profile of Figure 1 is the cleaner statement, and on it the conventional tail, not the atypical one, is the penalized end. The shape is the one audience-conformity research predicts for evaluators (Zhao et al., 2017; Hsu, 2006; Zuckerman, 1999), here from an examiner and at census scale; Taeuscher & Rothe (2020) find settings where the optimum sits off-centre.

3.2 Self-filed and professionally filed applications

Counsel of record is observed on every filing. Represented applications are 73% of filings over 1995–2018, falling from 85% in 1995 to 68% in 2018 as online self-filing spread. The two populations file differently and fare differently:

	Share of filings	Reached registration	Median terms (009 / 035)
Represented	73.3%	61.9%	28 / 26
Self-filed	26.7%	45.6%	21 / 17

Self-filers write shorter descriptions (a median of 21 terms against 28 in the software class, 17 against 26 in advertising and retail), sit closer to their class's usual language, and lean slightly further ahead of it. Because who drafts moves both the language and the outcome at once, counsel of record is held in every regression in the paper, and the later sections report their estimates separately by stratum. Appendix B draws the full flow from filing to registration split by drafter as a flow diagram (Figure 19).

3.3 Office actions and refiling: applicants rewrite toward the class

An application that is abandoned is usually the end of that attempt: among 2.28 million abandoned applications, 3.6% are followed within six years by the same owner filing the same mark again in the same class, and 1.5% by one that then registers. The refilings that do occur show what an applicant changes. Of the 42,115 abandoned applications whose owner later registered the same mark in the same class, 74% carry a different goods/services description the second time. Scoring each registration on the text it was first tried with and on the text it registered with gives the direction of the change. Both self-filers and represented owners rewrite toward the class's typical description: originals in the most atypical fifth lose 0.39 nats of atypicality, and originals in the most conventional fifth gain 0.16. Refilings that reuse the identical text move by less than 0.01 at every position, so the convergence is in the writing. Who drafts changes the extent but not the direction: owners who hire counsel for the second attempt rewrite most often (82%) and their descriptions lengthen by 16%; counsel rewriting their own earlier draft rewrite least often (72%) and shorten by 4% (Appendix C). Lead barely moves once the reference windows' shift with the later filing date is taken out.

Refiling after abandonment is itself mildly related to vocabulary position, though not on the axis registration responds to: refiling rates are flat in atypicality (a middle-minus-tails difference of -0.06pp on a 3.6% base) and rise with lead, from 2.6% among the most lagging abandoned applications to 3.4% among the most leading.

3.4 What the registration step selects

Three facts carry forward. Registration is a follow-through indicator, not an examiner's verdict: nine in ten failures are the applicant's own inaction. What examination itself rewards is a middling specificity — concrete enough to survive a definiteness objection, conventional enough to avoid a likelihood-of-confusion refusal — so the rate rises with atypicality and turns down only at the extreme, and is indifferent to lead, which no one at this stage could observe. And applicants who try again conform: rewritten refilings move toward the class's typical description from both tails, so the population of registered descriptions is compressed relative to what applicants first wrote. For the 42,115 refiled cases the first text is observed; for an applicant who conformed before ever filing it is not, and that part of the compression cannot be bounded. Every estimate on registered marks in the sections that follow is an estimate on this partly conformed language.

4 Persistence in commerce: the five-year proof of continued use

This section is about one outcome: whether a registered mark's owner proved, between the fifth and sixth year of the registration, that the product the mark names was still in commerce. The sample is every unique registration issued 2002–2018 that carries a score; failure is a dated cancellation for non-use at registration age 4.0–8.5 years (Section 2.6). Half of registered marks fail it. How the penalty found here moved across industries and decades is Section 5; what became of the firms is Section 6.

4.1 Why the analysis conditions on registration

The question is about companies whose products entered commerce: given that a product was brought to market, does the language of its first description predict whether it was still there five years on? Registration is the record's marker of entry — a use-based application asserts the mark is in commerce, and an intent-to-use application registers only after a sworn statement of use — so the analysis begins there. An application that never registered left the record, nine times in ten, by the applicant's own inaction, before any product met a customer. Including those applications would change the question into one about everyone who considered launching, including those who then did not, and since they are 2.3 million of the 6.0 million scored applications, their inclusion also buries the commercial signal in launch noise. Section 8.7 shows the pooled analysis for a reader skeptical of the conditioning: the registration step's shape is stamped onto every later outcome, and the gradients found below are attenuated and in places reversed.

Conditioning would miscount if abandonment were usually a pause rather than an ending — if owners reworked refused applications and refiled, the same product would be counted once as a failure and again as a success. It is not: among 2.28 million abandoned applications, 3.6% are refiled within six years and 1.5% lead to a registration (Section 3.3). Crediting every successful refiling as an eventual registration moves the registration profile by less than a tenth of a point. The refiled cases are 1.1% of this section's sample, they fail the five-year proof at nearly the sample rate (51.7% against 52.8%), and the headline contrast below is +2.25pp with them and without them.

4.2 Leading marks fail the five-year proof more often

Marks written in language their industry subsequently moved toward fail the five-year proof more often than marks written in language it was moving away from. Within Nice class and registration year, controlling for whether the filer drafted the mark themselves or hired counsel, and for the length of the description, registrations in the most leading fifth of lead fail 1.4 points more often (53.3%) than those in the most lagging fifth (51.9%; $t = 8.3$, Table 3).

Among 4.12 million registration class-records 2002–2018, dated cancellation at the five-year proof rises from 50.0% at the bottom lead quintile to 52.4% at the top, positive in 33 of 44 classes with sufficient volume (Figure 3). But a raw pooled contrast mixes the within-industry gradient with class and registration-year composition, treats multi-class registrations as independent observations, and ignores that outcomes correlate within owner. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations, one row per serial, with lead quintiles cut *within* Nice class and registration year. The specification adds class×registration-year fixed effects; controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis;[4] and standard errors clustered on 1.27 million normalized owners (Table 3).[5]

The gradient is monotone within cells. Whether the filing was drafted by counsel or by the owner is held because it moves both sides of the comparison at once: self-filers fail the five-year proof at 69.6% against 46.8% for represented owners, and Section 3.2 showed they write shorter, more

[4] A US application is filed either on *use* — the mark is already in commerce — or on *intent to use*, which defers proof until after allowance. Foreign applicants may instead file on a home registration (§44(e)) or extend an international registration into the US under the Madrid Protocol (§66(a)); the latter maintain under §71 rather than §8. These are statutory routes to the same registration, and they differ sharply in who files and how often the mark is abandoned, which is why each enters as a control.

[5] Ties are common. Within the class×registration-year cell the quintiles are cut in, 17.1% of scored registrations share a lead value with at least one other filing, and the largest tie group runs to 53. Anchoring the reference window on the filing date does not remove them, because filings that share a class, a filing date and a goods/services description — an owner registering a family of marks at once, on Manual language — have identical distributions scored against identical references. Cut points are fixed by sorting on the score and then on the serial number, which is reproducible but systematic: serials run in filing order. It makes no difference. Sorting the ties the opposite way moves the top-versus-bottom contrast by 0.001pp, and five pseudorandom orderings move it by at most 0.002pp, against a contrast of 2.287pp.

Table 3: Failure at the five-year proof of continued use and lead: design-based estimates. LPM of event-dated failure on within-class×registration-year lead quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class×registration-year fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3% on this sample.

Sample	Specification	Q5−Q1 (pp)	SE	n
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per sd of lead	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per sd of lead	+0.63	0.06	1,888,352

Standard errors replace significance stars: with n in the millions every coefficient has $p < 0.01$, so stars would mark each row identically. Errors are clustered on *normalized owner* — names uppercased, stripped of punctuation and of two dozen legal suffixes, so ACME CORP. and ACME CORPORATION are one cluster. A single owner files many marks, drafts them in a house style, and abandons them together when the business fails; the within-owner correlation of outcomes at the five-year proof is 0.38 pooled across classes (0.42 in the single-class checks where it was first measured), so the 1.27 million owner clusters, not the filings, carry the information. The Madrid row drops §66(a) registrations rather than absorbing them in a control, since they maintain under a different statute; the estimate is larger without them.

class-typical, slightly more leading descriptions, so without the control part of any lead gradient would be a self-filing gradient. With it, the penalty is of the same order in both strata (+1.8pp self-filed, +1.2pp counselled). The full control set cuts the magnitude by about a third without absorbing it, and the estimate is if anything *largest* in the cleanest stratum — counsel-represented, US-domiciled owners — so it is not carried by the self-filed or foreign mass-filing segments whose failure rates are high for their own reasons.[6] About forty percent of the raw pooled contrast is composition: +1.4pp design-based against +2.5pp raw. The penalty is also not constant over the sample: within counsel-represented US-domiciled owners the earliest third of cohorts carries none of it (−0.25pp, SE 0.33) — leading marks registered 2002–2007 were no less likely to survive than lagging ones — while the later thirds carry +2.00 and +2.65pp. Section 5 takes that variation as its subject.

Per class, the largest penalties sit in tobacco (+26pp, the vaping era), hand tools (+10pp), processed foods (+10pp) and medical apparatus (+9pp), with negative lifts in beer and soft drinks, transport and construction. These are the extremes of a 44-class screen of raw contrasts: the largest of many noisy estimates sit away from zero by selection, so the named classes locate where the penalty concentrates and their magnitudes should be read with that in mind.

[6]The level effects on those covariates are large in their own right: counsel −19.5pp and Madrid §66(a) basis +10.3pp.

4.3 The proof is a fixed appointment, and the penalty is a rate, not a timing

The five-year proof is close to a fixed appointment rather than a hazard spread over years: of the 1.35 million cancellations recorded for fully observed cohorts, 17 post before registration age 6.5, 96.1% post between 6.5 and 7.0, and the rest after seven. Since the cancellation record ends on 2 April 2026, a cohort is observable only once it has reached seven years, which makes 2019 the last one with usable data at this proof and leaves 2020 and later unobserved.

The outcome is an indicator on a window, so a mark that fails outside it counts as a survivor. If leading marks failed at the same rate as lagging ones but sooner, that alone would put them inside the window more often and the measure would report a difference in rates where there was only a difference in timing. It does not. On the 3.15 million registrations whose ninth anniversary is inside the record — for which failure is settled and no window is needed — the contrast is +2.71pp within class and registration year, against +2.65pp on the paper's 4.0–8.5 window. Conditional on failing, the probability of failing late carries a contrast of −0.01pp ($t = -0.2$) once class and registration year are held.

A competing reading is that the proof measures disengagement rather than products: an owner who has lost interest both answers the examiner slowly and lets the registration lapse. The prosecution record tests this through the gap between the first non-final office action and the response to it. Response speed does not predict the outcome: across the 1.22 million scored registrations with a usable office-action/response pair, passage varies between 46.1% and 48.1% across response-speed quintiles, from a median of seven days to 183, and the quickest responders pass slightly *less*. The lead penalty is present in both halves, stated on passing: −3.48pp (±0.40) among faster-than-median responders and −2.36pp (±0.40) among slower, against −2.94pp pooled. The restriction drops 61% of the sample, disproportionately applications that never drew a refusal, so this speaks to the prosecuted subset.

4.4 The penalty under three theme scorings

Every estimate above scored filings on fifty themes fitted once over the whole corpus, with each class supplying its own reference. That is one of three ways to partition the vocabulary, and the penalty does not depend on the choice. The identical registrations were re-scored under *500 global themes* (the same construction at ten times the resolution, so a theme is closer to a product category than to an industry) and under *50 themes fitted per class* (a separate model on each class's own filings, so nothing is shared across classes). In all three, surprise is measured against the filing's own class on per-filing windows of 1,826 days each side.

The lead penalty is a property of the filing, not of the partition (Figure 4, left): +2.29pp under the production scoring, +2.61pp at 500 themes, and +1.94pp under per-class themes, each with a standard error of 0.09, and positive under all three at once in 28 of the 44 classes with at least 3,000 sampled registrations. The scorings agree only moderately filing by filing — lead correlates at 0.64 between the two global resolutions and at 0.52 between the global and the per-class fifty — so a third to a half of the variance in any one scoring's lead is particular to that scoring, and the penalty survives the disagreement. (Table 4 tabulates the contrasts; the standard errors here are two-sample binomial and unclustered, and outcomes correlate within owner at 0.38, so single-class t-statistics overstate precision by roughly a factor of two.)

Atypicality behaves differently (Figure 4, right). Under the two global scorings the most atypical fifth of registrations fails the five-year proof 4.3–4.4 points less often than the least atypical; under per-class themes the gap is 0.7 points. The two global scorings' atypicality correlates at 0.79; global and per-class atypicality correlate at 0.39. A global model places every class in one set

Table 4: Failure contrasts at the five-year proof under three theme scorings, on the identical registrations. Top-minus-bottom quintile, quintiles cut within Nice class and registration year, no controls; $n = 3{,}105{,}048$, base failure 52.3%.

Scoring	Lead Q5−Q1 (pp)	SE	Atypicality Q5−Q1 (pp)	SE
Global, 50 themes	+2.29	0.09	−4.39	0.09
Global, 500 themes	+2.61	0.09	−4.29	0.09
Per class, 50 themes	+1.94	0.09	−0.72	0.09

of coordinates, so a filing that imports a theme the class rarely uses and other classes use often registers as atypical for its class. A model fitted on the class alone re-partitions the class's own vocabulary, and the imported theme becomes a local theme like any other. What the global scoring calls atypical is, in large part, drawing on a theme from elsewhere.

That population can be measured directly, and it separates two kinds of new. Under the 500-theme model, for each theme and class, the first sustained year is the first year by which the theme has been the dominant theme of at least one hundred filings in that class; the corpus-wide first sustained year is the earliest of those across classes.[7] Each registration in the sample is assigned its dominant theme and classified at its filing date as carrying a theme new to the corpus, a theme new to its class (established elsewhere, arriving here), or an established theme. Of 4.12 million registration class-records, 3.97 million carry an established theme, 151,012 a theme new to their class, and 779 a theme new to the corpus — the last too few to support a claim, because almost every theme a 500-theme model can resolve had been established in some class before 2002.

Registrations carrying a theme new to their class fail the five-year proof at 48.3% against 51.6% for established themes, and 0.43 points less often within class and registration year (SE 0.13). Holding the filing's own lead and atypicality as well reverses the sign: net of them, the new-to-class registrations fail 1.06 points more often. The two statements describe one population. A theme arrives in a class in filings that are atypical for that class, and atypical filings survive; that is the −0.43. Among filings of equal atypicality, the ones carrying the arriving theme fail more; that is the +1.06. The atypicality protection of the global scorings is therefore mostly the protection of extension: language established elsewhere and extended into a new class survives. The penalty attaches to being early with language, not to being far from the class's centre.

Nor is the lead penalty an atypicality effect in disguise. The two could be confounded mechanically — lead is the difference of two nonnegative levels whose average is atypicality, so its spread grows with atypicality and its magnitude correlates with it at +0.29 — but the signed value does not (−0.05), and cutting lead quintiles within fifths of atypicality leaves the penalty intact in every fifth: +2.1, +3.1, +3.3, +1.7 and +2.5pp from the least atypical fifth to the most (each SE under 0.3).

4.5 The risk persists at the year-ten renewal

Survivors of the five-year proof face a combined declaration and renewal at year ten. On the design-based estimates the lead penalty does not attenuate there: lead enters at −0.49pp per standard deviation on passing the five-year proof and −0.63pp on renewing at year ten (Table 10; both stated on survival, so the signs run opposite to the failure convention above). What attenuates is the raw single-class contrast: in the software class, where the five-year lift on the 2010–2013 cohorts reaches

[7] Sustained years and theme shares are computed on a 25% systematic sample of all filings (every fourth serial), scaled up, over filing years 1986–2024.

+12.1pp, the conditional lift at the renewal falls to +7.6pp. Pooled across classes the renewal profile is not a slope at all. Fitted over 872,079 five-year survivors with a resolved year-ten status, a straight line through renewal against lead explains 1% of the variation across forty bins and a V explains 89% (ΔAIC = 83). Renewal is highest just below the centre of the lead axis (64.2% near $z = -0.4$) and lowest at both tails: 45.5% at the most lagging bin — the lowest anywhere on the axis — and 50.6% at the most leading. Most of the difference between leading and lagging resolves at the five-year proof; what remains at renewal is a penalty on both extremes of the lead axis, the lagging one the larger.

4.6 What the five-year proof shows

Among products that entered commerce, being described in language the industry was moving toward predicts being gone within six years: +1.4 points on a 52.3% base with the full design, +2.3 raw, in three of every four classes, under every partition of the vocabulary, robust to timing, to drafting, and to who files. Atypical language predicts the opposite, but that protection belongs mostly to filings extending a theme established elsewhere into a new class; net of atypicality, even those arriving-theme filings fail more. The market's step and the examiner's step read the text differently for their own reasons: the examiner reads distance from the class's usual language at a moment when lead does not yet exist to be read (Section 3.1); the five-year proof is recorded after the industry's language has moved, and it is where being early shows up as a cost.

5 New industries, fads, and waves

This section asks when and where the leading penalty of Section 4 lived: across registration cohorts, across the technology classes, across the themes each era's leading filings were drawn from, in the filings that name the internet, and inside surging themes. The outcome throughout is the same one — failure at the five-year proof of continued use — and every sample is settled cohorts (registrations whose ninth anniversary is inside the record), so nothing here is censored.

5.1 The penalty weakened in the middle of the sample and was largest at the start

Scores exist from 1995, which makes registration cohorts from 1996 evaluable, and their outcomes at the five-year proof have been settled for two decades. Figure 5 draws, for each cohort, the change in survival for a one-standard-deviation increase in lead and in atypicality, the two entered jointly and standardized within Nice class and cohort. Pooled across classes, the lead coefficient is significantly negative in every settled cohort except 2002–2005 — −1.9 to −2.3pp per standard deviation for the 1996–2000 cohorts, +0.4 to −0.2 across 2002–2005, and −0.5 to −1.4 from 2006 on. In the software class the middle of the sample is not a pause but a reversal: the coefficient runs from −6.8pp for the 1996 cohort to significantly *positive* (+0.9 to +1.2) in 2002, 2003 and 2005, then back through −4.0 in 2012. The atypicality coefficient never changes sign anywhere: more atypical registrations survive more in every cohort of both panels, by +0.6 to +2.5pp per standard deviation pooled and +2.9 to +5.7 in software.

Stated as cohort contrasts, the swing is large: +6.9pp excess failure for the most leading fifth in the 1996 cohort, rising to +10.1pp for 1999 — roughly twice the largest cohort contrast observed since — then collapsing, and returning to +1.8 to +5.6pp for every cohort from 2008. Scores begin in 1995, so a 1996-cohort registration must have issued within a year of filing; early cohorts are selected on prosecution speed. Holding pendency in a common one-to-three-year band leaves the series unchanged. Indexing by filing year instead, so a cohort contains every prosecution speed, gives +7.9 to +9.6pp for filings 1995–1999 and −0.95pp for filings made in 2000 — a ten-point swing in a single filing year.

Splitting the four classes that carry most technology filings — 009, 035, 038 and 042 — against the other forty-one separates a large moving component from a small steady one (Figure 6):

Filing years	Technology classes	All others
1995–1999	+14.18 (0.36)	+4.48 (0.27)
2000–2004	−1.37 (0.35)	+1.21 (0.24)
2005–2007	+2.04 (0.43)	+1.99 (0.28)
2008–2014	+9.02 (0.25)	+1.32 (0.17)

Outside technology the penalty is a flat +1 to +4pp in every era, including the one in which the pooled series reads as zero. Inside technology it swings by fifteen points: strongly positive for filings made into the late 1990s, negative for those made in the five years after, and positive again from 2008. The pooled quiet of 2002–2005 is a technology reversal partly offsetting a non-technology positive, not an absence.

What this design cannot say is which cycle the swing belongs to. Marks filed 1995–1999 reach their five-year proof in 2001–2007 and marks filed 2000–2004 reach theirs in 2006–2013, so a story in which unusual language is punished because it was written during an investment boom and a story in which it is punished because the proof falls due during a bust both fit these numbers.

5.2 Three of the four classes reverse, and the leading language changes theme

The four-class split pools industries that do not move together. Taking them one at a time, and then assigning quintiles inside progressively finer cells (Table 5):

Table 5: The leading penalty at the five-year proof inside technology, by class and by cell. Top-minus-bottom lead-quintile contrast in failure (pp, SE in parentheses), settled registrations, by filing era. The upper block assigns quintiles within each class and era. The lower block pools the four classes and assigns quintiles within the era (raw), within class×registration-cohort cells, and within theme×class×registration-year cells, where a filing's theme is the one its description loads most heavily on under the production $T = 50$ model. n = 1,248,591.

Class	1995-1999	2000-2004	2005-2007	2008-2014
009 Electrical, software	+18.04 (0.52)	−5.32 (0.51)	+0.25 (0.62)	+11.67 (0.37)
035 Business services	+4.59 (0.69)	−9.70 (0.56)	+2.02 (0.63)	+1.66 (0.36)
038 Telecommunications	+9.91 (1.40)	−4.24 (1.29)	+2.60 (1.48)	+3.42 (0.94)
042 Scientific, computer services	+17.11 (0.57)	+6.70 (0.62)	+8.42 (0.87)	+12.63 (0.47)
Four classes pooled, raw	+14.49 (0.33)	−1.81 (0.31)	+2.56 (0.38)	+9.16 (0.22)
within class×cohort	+14.27 (0.33)	−3.63 (0.31)	+2.73 (0.38)	+7.24 (0.22)
within theme×class×cohort	+4.78 (0.33)	−1.26 (0.31)	+2.56 (0.38)	+4.51 (0.22)

Electrical and software goods (009) and telecommunications (038) follow the pooled pattern. Business services (035), the class that holds online retail, reverses hardest: −9.7pp for filings made 2000–2004 against +4.6pp in the five years before. Scientific and computer services (042) never reverses. Its contrast is +6.7pp in the era in which the other three are negative and between +8 and +17pp in every other, and by registration cohort it never falls below +3pp (Figure 7). The reversal is an event in goods and in business services; the class that sells technology as a service sat it out.

Assigning quintiles within class and registration year rather than within the era pool changes little. Assigning them within theme, class and registration year changes a great deal: the late-1990s contrast falls from +14.5 to +4.8pp, the post-2008 contrast from +9.2 to +4.5pp, and the 2000–2004 reversal from −3.6 to −1.3pp. Two thirds of the first and half of the second are between themes rather than within them. The leading fifth of each era was drawn from particular themes, and those themes carried their own failure rates; the penalty for being leading *inside* a theme is a steadier +2.5 to +5pp, the same order as the penalty outside technology.

Which themes is a matter of record (Table 6; the table's last row is the all-themes pooled contrast within theme×class×cohort). In 1995–1999 the computer, software, information and data services theme held 38% of technology filings and 75% of the leading fifth, and its registrations from those filing years failed at 64.9% against 47.5–59.0% in the other large themes. That one theme is most of the late-1990s penalty. In 2000–2004 it held 12% of the leading fifth. The leading language had moved to video, music and games on electronic media (31% of the leading fifth against 12% of filings) and the internet-services language had become lagging — wording the class was leaving — and inside several themes the lagging fifth now failed more than the leading one: −3.6pp in electronic media, −5.5 in retail store services, −6.9 in apparatus and control systems. From 2008 the leading fifth over-represents downloadable and online software by four to one (15% against 4%), the vocabulary of the application store; those registrations fail at 67.2% over the whole period, the highest base rate of any theme, and their within-theme lead contrast pooled across cohorts, +8.4pp, is the largest of any theme (the table's 2008–2014 cell, cut within that era alone, is +3.3pp). The computer-services theme's own contrast is positive in every era (+3.6 to +5.9pp) and never reverses.

Table 6: The leading penalty at the five-year proof by theme within the four technology classes. Themes with at least 20,000 settled registrations, labelled by their five highest-probability terms; quintiles assigned within theme and era. Cells with fewer than 3,000 registrations are blank. The final row pools all themes, with quintiles cut within theme×class×cohort.

Theme (top words)	n	1995-1999	2000-2004	2005-2007	2008-2014
services, computer, software, information, data	451,253	+3.61 (0.52)	+3.56 (0.49)	+4.18 (0.64)	+5.92 (0.37)
video, computer, electronic, music, games	162,290	+2.53 (0.97)	−3.62 (0.89)	−2.85 (1.01)	+3.36 (0.60)
services, featuring, store, retail, store services	131,107	+8.39 (1.09)	−5.49 (1.01)	+2.92 (1.15)	+0.98 (0.66)
services, field, training, estate, educational	95,602	+7.82 (1.09)	+0.99 (1.12)	+6.02 (1.43)	+2.05 (0.83)
apparatus, systems, devices, electronic, control	84,121	+4.80 (1.27)	−6.92 (1.19)	−0.58 (1.41)	+3.65 (0.88)
design, field, systems, development, services	32,517	+10.40 (2.03)	+4.67 (1.82)	+1.50 (2.27)	−2.27 (1.45)
software, online, downloadable, non-downloadable, virtual	29,472	—	—	—	+3.30 (1.05)
bags, clothing, shirts, jackets, leather	28,421	+1.28 (2.59)	−1.55 (2.39)	−4.66 (2.40)	+6.06 (1.36)
research, medical, scientific, testing, purposes	25,897	+12.65 (2.66)	+12.09 (2.03)	−3.82 (2.59)	+5.63 (1.51)
services, health, field, care, information	22,880	+4.94 (2.04)	+1.39 (2.36)	—	+3.50 (1.72)
All themes, within theme×class×cohort	1,150,921	+4.78 (0.33)	−1.26 (0.31)	+2.56 (0.38)	+4.51 (0.22)

So the swing has two layers. Across the whole period, a leading filing is a few points more likely to fail than a lagging one in the same theme, class and cohort, in technology as elsewhere. On top of that, each era's leading vocabulary was concentrated in one or two themes whose marks failed at their own rate: internet services in the 1990s, application software after 2008. In the trough between, the language that had been leading became lagging, and the marks still written in it failed with it. The design-based estimates of Table 3 hold class and cohort fixed but not theme, so they carry both layers; a reader who wants only the within-theme penalty should take the lower block of Table 5 as the guide to its size.

5.3 Internet as a theme: web-based filings against their class

A filing is internet-bearing when its goods/services text names the internet, the web, a website, online or on-line provision, or electronic commerce. The rule reads the text rather than the class, so a web retailer filing in apparel is caught and a semiconductor maker filing in class 009 is not. Internet-bearing filings are 16.9% of scored registrations: 58% in telecommunications, 39–41% in advertising and retail, scientific and technical services, and legal and personal services, under 5% in most goods classes. Figure 8 puts every class on two axes at once — how much more often its internet-bearing filings register, and how much more often they fail the five-year proof — and Table 7 carries the underlying rates.

Three patterns are in the figure. In the industrial goods classes, internet-bearing registrations fail the five-year proof far more often than their class: machinery 50.9% against 38.7%, chemicals 46.1% against 39.1%, vehicles 49.6% against 45.1%. A web-based business registering in an engineering class fails at the rate of a software registration (51.4%), not at the rate of its class. In the consumer goods classes the sign reverses — internet-bearing registrations in clothing fail 52.4% against 58.7% for the class, in household utensils 44.5% against 50.5% — so online sellers of consumer goods outlast the class's other registrants. In every service class the internet-bearing filings fail more often, by one to eleven points, most in legal and personal services (60.1% against 48.7%). At the registration step the industrial-goods classes sit to the right of zero: internet-bearing applications there reach registration 59.5% of the time against 56.0% for the rest. For those classes the two steps move in opposite directions — more likely to be granted, then more likely to be cancelled for non-use six years later — which is the pattern Section 4.6 predicts when the examiner reads distance from class language and the market reads whether the product stayed in commerce.

The failure gap has a common shape in event time. Dating the internet's arrival in each class as the first filing year in which the pattern reaches 2% of the class's filings (1994–1995 in the technology and transport classes, 1999–2000 in apparel and chemicals, 2008 in machinery), and aligning each class's per-cohort gap on years since its own arrival: the gap starts near +7 points in the first observable cohorts, declines to about zero nine to eleven years after arrival, and settles between +1 and +4 (Figure 9). Across the eight classes observable in both phases, the mean gap is +4.4 points in years zero to four and +1.6 from year ten, shrinking in six of the eight. Calendar time disguises this: transport, whose arrival was 1995, peaks at +16 points in its 2000 cohort and declines; machinery, whose arrival was 2008, peaks at +17 points in its 2012–2015 cohorts — its own early years. The class's clock starts when the vocabulary arrives, not when the internet did. Clothing is the standing exception: its internet-bearing registrations outlast the class in every cohort after 2004, consistent with online sellers of goods surviving where early web-services ventures did not.

One reading of the early-cohort excess is entry cost. Opening a web-based business in a class's first internet years required little capital and little prior experience in the trade, while capital for exactly such ventures was abundant; a step that filters on whether a product held its place in commerce for five years will fail cheap entries at a higher rate wherever they cluster, and they clustered in each class's first internet cohorts. The record cannot test founders' costs directly, but the event-time shape — highest at each class's own arrival, fading as the channel became ordinary — is what falling entry barriers followed by normalization would leave.

The internet split also locates the era swing of Section 5.1. Rerunning the per-cohort lead coefficients of Figure 5 separately for internet-bearing and other registrations (Figure 10): the mid-sample reversal belongs to the internet-bearing filings, whose leading registrations survived *more* in the 2001–2005 cohorts (+1.3 to +2.5pp per standard deviation) while non-internet technology registrations sat near zero. The other two episodes are not the internet's in this textual sense. The late-1990s penalty is present in technology registrations that never name the web (−3.4 to −4.8pp for the 1996–2000 cohorts) — consistent with Table 6, where that era's deadly vocabulary is computer and software services, words that predate explicit internet naming. And the post-2008 penalty sits in both splits at once (−2 to −3pp each). Outside technology, the non-internet series is the steady −0.4 to −1.4pp throughout. What the internet changed, on this cut, is the middle: for half a decade after the crash, the web filings whose language led their class were the ones that lasted.

The classification system has faced this question and twice declined to make the internet a class. Before 2002, internet and computer services accumulated in class 42, whose heading then ended "services that cannot be classified in other classes"; the Nice Union's Committee of Experts restructured it in the eighth edition, effective 1 January 2002, by narrowing 42 to scientific and technological services and moving the rest into three new classes (43–45), organised by the kind of service rather than by the technology.[8] The question returned with virtual goods: the twelfth edition, effective 1 January 2023, classifies downloadable virtual goods and NFT-authenticated files in class 9 and distributes metaverse-related services across the existing service classes, again declining a new class.[9] The pattern above bears on that choice. Internet-bearing filings in a goods class survive or fail at rates closer to the technology classes than to their own class, so for the outcome this paper measures, the technology overlay carries more information than the class assignment it is filed under. A classification built for scope of protection has no reason to track survival; but a reader using Nice classes as industry controls, as this paper and the literature do, should know that the internet cuts across them.

5.4 Surges: joining a wave

Genuinely new themes are rare (Section 4.4); the observable event is a surge, an established theme's share doubling, over its own three prior years, to a level that matters. Figure 11 draws four episodes: their share trajectories, and how their registrations fared at the five-year proof.

Measured against the whole corpus at a 1% share, five themes ever surge, and one episode clears a 1,000-registration floor: the online-and-downloadable-software theme doubled to 2.4% of all filings in 1999. The 6,339 registrations whose dominant theme it was during 1999–2001 failed the five-year proof at 63.7%, 8.6 points above their class and year and 8.5 net of their own lead and atypicality. Within the episode, the most leading fifth failed no more than the most lagging (−0.2pp, SE 1.9).

Measured within a class at the same thresholds, surges are common: 962 episodes of a theme doubling to at least 1% of its class, holding 32,418 registrations at the five-year proof. Pooled, being in one of these smaller waves costs little: in-surge registrations fail 0.65 points more than

[8] Nice Classification, 8th edition (WIPO, 2002); the USPTO's guidance describes the class-42 restructuring. The eighth edition took effect in the middle of the era swing of Section 5.1.

[9] Nice Classification, 12th edition (WIPO, 2023).

their class and year, 1.27 net of lead and atypicality (1.33 and 1.70 at the 2% threshold). The episodes differ in sign among themselves, as the figure's right panel shows: the education-services surge in advertising and retail after 2002 failed 13.7 points less than its class; the charitable-fundraising surge in finance after 2011 failed 4.0 points more. Of the few episodes large enough to carry their own lead contrast, one is clearly positive (the education-services surge, +5.8pp, $t = 2.2$) and the rest are indistinguishable from zero.

Three further facts about the waves, on the 61 episodes with at least 300 joining registrations over a six-year entry window — 64,105 of the 86,447 registrations that join any episode. First, within a wave, entry order does not matter: demeaned within episode, joiners in the surge year fail −0.8 points relative to their wave's mean, joiners five years in +0.0, with no gradient between (each SE about 0.5). Second, among class-scale waves, larger is not deadlier: excess failure is +2.7 points in the smallest tercile of waves, +3.9 in the middle, −0.1 in the largest, and correlates with wave size at −0.22. The eight-point excess of the corpus-scale software wave is not the top of a dose-response within classes; it is a different scale of event. Third, call a wave growing when the theme's share five years after the surge is at or above its surge level. Growing waves carry more excess failure (+4.3 points, 22 episodes) than receding ones (+1.0, 39 episodes). That runs against reading the excess as pure fad mortality, in which the collapsing waves should be the deadly ones, and against under-response, in which joiners of durable waves should thrive. A reading consistent with all three facts is selection at the proof rather than crowding out: a surging theme draws registrations for products that were never brought to use at higher rates wherever the theme is commercially active, and the theme's own subsequent growth does not rescue them. With 22 growing episodes whose individual excesses range from −13.7 to +4.0 points, the 3.3-point difference between growing and receding waves is suggestive, not settled.

5.5 What the waves show

The leading penalty is steady and small outside technology and episodic inside it, and most of the episode is between themes: each era's leading vocabulary was concentrated in one or two themes that failed at their own rate, while the within-theme penalty holds at +2.5 to +5pp throughout. Web-based filings fail hardest in each class's own first internet years and converge toward their class within a decade. Joining a surging theme costs little on average and nothing for arriving early within the wave; the one corpus-scale wave, online software in 1999, is the exception at every margin. What a fad costs, on this record, is mostly a matter of which theme the fad was, not of when the filer joined it.

6 Commercial success: funding, listing, and the firms that mattered

A mark that survived the five-year proof names a product that held its place in commerce. This section asks whether the owners of leading or atypical debut filings went on to the outcomes that make a firm rich: a priced private round, SEC financial reporting, a public listing, and revenue at scale. The unit is the owner's first registered filing, its debut, scored on the production measure; the outcomes are dated events in the SEC record. Form D became mandatory in electronic form on 16 March 2009, so the funding ladder is built on debuts made 2009–2018, for which every rung is observed; the public-reporting analysis uses the longer 1995–2018 sample.

6.1 The ladder: a round, reporting, a listing

Of 999,442 owners whose first registered filing, carrying a score, was made 2009–2018, 17,115 (1.71%) raised a priced private round after the debut, 4,383 (0.44%) appear in SEC reporting records at any time, and 1,795 (0.18%) registered a class of securities or completed an initial public offering at any time. The funded rung requires the round to postdate the debut; the reporting and listing rungs do not, so owners already reporting before their first trademark filing are counted, and the staged analysis in Section 7 shows they are a majority of the exchange-listed. The rungs nest: 7.3% of funded owners reach reporting against 0.3% of the unfunded, and 41% of reporting owners reach an IPO.

Table 8 gives the rate at each rung by fifth of lead and of atypicality, with fifths cut within Nice class, debut year, and fifth of description length, so that neither the industry, the year, nor the length of the description enters the contrast. Funding does not vary with either axis: the most leading fifth of debuts is funded at 1.80% against 1.86% for the most lagging, and the most atypical at 1.64% against 1.68%, both inside sampling error. Trademark counts and breadth carry a funding signal (Block et al., 2014); vocabulary position does not. Reporting and listing do vary. Owners whose debut language was most lagging reach reporting at 0.59%, owners whose debut language was most leading at 0.38%; for an IPO the rates are 0.24% and 0.17%. On atypicality the sign is positive on this sample: the most atypical fifth reports at 0.53% against 0.42% for the most conventional, and lists at 0.22% against 0.15%. Section 6.5 absorbs description length continuously on 1995–2018 debuts and finds the atypicality gradient flat; with length held in fifths on 2009–2018 debuts it is positive at t above 4. The two are stated side by side; the difference between them is a length control and a sample, and the atypicality result at the listing margin is the one claim in the paper that depends on which is used.

The lead profile at reporting and at IPO is not monotone: it falls from the lagging end through the fourth fifth and turns up in the fifth. Owners whose first description used language the class was leaving behind are the likeliest to report and to list. Most of them are established firms making a first trademark filing late in a business already built on the class's older vocabulary, and the ladder cannot separate that from anything the language itself does.

6.2 Listings without an observed private round lean lagging

Of the 1,795 owners with an IPO marker, 807 have no post-debut Form D round in the record. Their debut language leans lagging: mean lead, standardized within class and debut year, is -0.25 (SE 0.04) against -0.08 (SE 0.04) for the 988 who raised an observed round, and 29% of them sit in the most lagging fifth against 22% of the funded. Both groups are mildly atypical (+0.13 and +0.14 standard deviations). The unfunded route to a listing belongs disproportionately to owners whose first description used language its class was moving away from — consistent with established, revenue-financed businesses listing without venture money — though "unfunded" here means only that no Form D notice is observed: rounds before 2009, foreign placements, and registered offerings do not post one.

6.3 The firms that produced the most value

Among the 1.80 million owners whose first registered filing was made 1995–2018, 4,710 can be linked to SEC financial statements with reported revenue, and their peak annual revenues sum to \$19.2 trillion. Ten of them hold 18.5% of that total, fifty hold 41.5%, five hundred hold 82.9%. Reading this handful of owners is not a statistical argument — no gradient estimated on 4,710 firms carries the paper's weight — but it is not a stray anecdote either: these are the firms in which most of the linked economy's value was generated, and the record shows what their first filings looked like.

Their debut language was ordinary. Standardising lead and atypicality within class and debut year, the ten largest firms' first registered descriptions sit 0.35 standard deviations below the average in atypicality and within 0.01 of it in lead; the fifty largest sit 0.08 below in atypicality and 0.13 above in lead; the five hundred largest are within about 0.1 of the average on both. Weighting every linked firm by its revenue, the linked set as a whole sits 0.02 standard deviations below the average in atypicality and at the average in lead. The firms that produced most of the value described their first product in language their industry already used, and no more ahead of it than the rest.

6.4 Internet, blockchain, and artificial intelligence

Three vocabularies were hand-curated from the goods/services text: internet (the internet, the web, websites, online provision, electronic commerce), artificial intelligence (artificial intelligence, machine learning, deep learning, neural networks, natural language processing and understanding, computer vision, predictive analytics, chatbots, generative AI, large language models), and blockchain (blockchain, cryptocurrency, crypto assets and tokens, bitcoin, non-fungible tokens, distributed ledgers, digital and virtual currency, smart contracts). Counted as class-filing rows over the full corpus, internet vocabulary appears in 2.94 million rows, AI in 167 thousand, blockchain in 114 thousand. Registration rates differ: 52.1% of internet filings registered, 34.5% of AI filings, 31.2% of blockchain filings, against 57.6% of all debut filings; the last two are recent and many are still in prosecution or abandoned before use.

On the 2009–2018 debut ladder the three behave alike in one respect and differ in another. AI debuts are funded at 7.8% and blockchain debuts at 8.4%, against 2.9% for internet debuts and 1.7% for all debuts: a first filing in AI or blockchain vocabulary is four to five times as likely to be followed by a priced round. Reporting and listing do not scale the same way: AI debuts report at 0.66% and list at 0.38%, blockchain at 1.08% and 0.36%, internet at 0.52% and 0.22%, against 0.44% and 0.18% overall. The funding gap is a factor of five; the listing gap a factor of two, on small counts (1,824 AI debuts, 1,390 blockchain).

6.5 Leading debuts reach public reporting less often; unusual ones no more often

The outcome here is whether an owner ever appears in the Securities and Exchange Commission's financial reporting records. That is broader than exchange-listed: roughly half the matched firms carry a ticker and the rest report for other reasons. Figure 12 draws the raw rate against each axis in twenty bins. In signed lead the profile is a U whose tails both sit above the base rate and whose lagging tail is the higher; in lead magnitude and in atypicality the raw rate rises.

The raw atypicality gradient does not survive the design-based estimates (Table 9). Within class and filing year, holding description length fixed, top-quintile atypicality raises the probability of SEC reporting by +0.000pp on a 0.481% base ($t = 0.0$), and by −0.003pp on the past level alone. The quintile profile is not even monotone: the three middle quintiles sit *below* the bottom one and the top merely returns to it. The 2.7-fold gradient in the raw bins is composition: firms that go on to report cluster in particular classes, years, and multi-class filings, and class×year fixed effects absorb it. Lead costs, at equal atypicality: the signed component enters negatively (−0.047pp per standard deviation, $t = -7.8$) while the unsigned component enters positively (+0.067pp per standard deviation, $t = 7.7$). The U in the raw bins is exactly this pair: lagging filings show the magnitude premium with no lead penalty, leading filings show it net of that penalty, and the middle bins show neither. Once the levels are controlled, the recovery of the far-leading tail is much reduced: the quintile profile falls through Q4 and recovers only slightly at Q5.

Owners are matched to SEC registrants by normalized exact name, over a candidate universe of all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, and one match is kept per owner string. That yields 19,889 matched owners covering 7,588 registrants, and a base rate of 0.481% among registered debut filers. Validation on 22 companies that indisputably went public — across software, pharmaceuticals, food, apparel, aerospace and autos — matches all 22. A random sample of 25 matches carrying at least three filings was inspected by hand and none was wrong, which is a check on gross failure modes rather than a precision estimate.

The linkage is nonetheless a name match, and that bounds the atypicality reading in particular. Firms with distinctive names are easier to match than firms with generic ones, and distinctiveness of a name plausibly travels with distinctiveness of the goods/services text; nothing in this design separates the two. The lead coefficient is unchanged under alternative linkages; the atypicality coefficient is not, so only the lead result is robust to the name-match design. The financing rows of Table 10 rest on a Form D match whose raw extracts are not in the current tree and should be read as provisional.

The U is also not atypicality working through lead's construction. Lead magnitude and atypicality correlate at +0.25 on this sample — a filing near the typical core cannot carry a large lead either way — so a gradient attributed to the signed axis could in principle be an atypicality gradient in disguise. Signed lead and atypicality correlate at only −0.10, and cutting lead fifths within fifths of atypicality keeps the structure: the most lagging fifth out-reports the most leading in four of the five atypicality strata (by 0.08 to 0.39pp on sub-1% bases), and the ordering reverses only in the most atypical fifth, where the most leading debuts report most. The lagging tail's premium is not a ceiling artifact; the leading tail's partial recovery in the raw U is where the atypicality composition lives, which is what the level controls in Table 9 absorb.

"SEC reporting" is not "went public": splitting matched owners on whether their SEC filer number carries a ticker gives 10,556 listed against 9,333 reporting without one, and both halves show the same U in lead (Section 8), so the result is not an artifact of counting non-operating registrants as successes.

6.6 The construct is not patenting

Within the firms that do both, trademark vocabulary and patenting move independently. On a panel of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024), each axis is regressed on $\log(1 + \text{patents})$ with firms absorbed and errors clustered on the firm, estimated on 94,181 firm-years across 17,506 firms under theme scoring (93,551 and 17,415 under term scoring) — 134,075 matched firm-years before dropping the firm-years that contribute no within-firm variation. The outcome is standardized, so β is the shift in standard deviations of the axis per unit of log-patents; one unit is roughly a tripling of the patent count, and a firm going from none to twenty moves about three units.

Text counted as	Axis	β (sd)	t	Between-firm r
Themes	atypicality	+0.012	+3.0	—
Themes	lead	−0.058	−9.4	−0.020
Terms	atypicality	−0.055	−11.5	—
Terms	lead	+0.012	+2.0	—

The grid is close to a mirror: whichever axis carries the small signal under one way of counting the text carries the opposite sign under the other. The theme-scored *lead* coefficient (-0.058) and the term-scored *atypicality* coefficient (-0.055) are nearly the same number attached to different quantities. Only those two cells are established; the mirror's other diagonal is within noise, and the orthogonality conclusion rests on the magnitude bound — every cell under 0.06 standard deviations — rather than on any cell's sign. Nothing here reaches a tenth of a standard deviation.

One objection is timing: a patent is applied for before its invention is in public use and a trademark when the firm is ready to sell, so patents would lead. Firms do not proceed that linearly, and in some industries the trademark comes first (Seip et al., 2018). Entering patents at $t-2$, $t-1$, t and $t+1$ — the last allowing the trademark to precede the patent — one offset at a time, since the count is strongly autocorrelated within firm, gives -0.054 to -0.089 standard deviations of log patenting under theme scoring and $+0.012$ to $+0.024$ under term scoring. No ordering isolates a direction. Among the 8,760 owners on the patent panel who also raised a priced round, the first round came before the first patent grant for 49.9%, after it for 41.8%, and in the same year for the rest; the median lag is zero years. Neither event predicts the other's timing.

The second objection is heterogeneity: a pooled near-zero is consistent with genuine orthogonality everywhere, but equally with a positive slope in sectors where the two rights are complements — pharmaceuticals, chemicals, medical devices, machinery — offsetting a zero or negative slope where trademarks do the work alone. Testing that requires sorting classes into complements and substitutes, and the sorting available is coarse and judgement-based: classes were assigned in advance from their definitions alone — the eleven covering chemicals, pharmaceuticals, machinery, electronics, medical apparatus and scientific services as complements, the consumer-goods and services classes as substitutes, the rest neither — with nothing estimated and no validation against observed joint filing behaviour. On that sorting (a sectoral build scored at 200 themes, firms with at least three panel years, 50 firms and 300 firm-years per cell), the slopes are -0.000 ($t=-0.03$) among complements, -0.026 ($t=-1.45$) among substitutes, and an interaction of -0.008 ($t=-0.49$). Across the 27 Nice classes clearing a 50-owner floor, the complement classes run from sixth place to twenty-fourth by coefficient and do not agree in sign with one another; the only significant cells are paper and housewares, both negative and neither a complement class. Neither retention supports the complementarity story, with the caveat that a coarse sorting can miss a real but finer-grained one. The test is also conditional on a successful assignee match, so every owner patents at least once and the cells resemble each other more than the industries they stand for do, pushing any interaction toward zero.

6.7 Leading does not pay, even where novelty does

The answer to this section's question is no. An owner whose surviving product had been described ahead of its industry's language is not more likely to raise a priced round, reaches SEC reporting and listing *less* often than a lagging owner, and is absent from the top of the linked revenue distribution, whose firms debuted in ordinary language. What pays at the reporting margin is unsigned: lead magnitude and description length, the marks of a specific, seriously drafted first filing, carry positive coefficients while the signed component stays negative. Being unusual is at worst free and at best mildly rewarded; having been early is never rewarded at any rung the record shows, and the listings that arrive without observed venture money belong disproportionately to the lagging.

7 Discussion: each axis bites at a different stage

Table 10 re-estimates every outcome on one harmonized sample — one row per debut owner, scored on that owner's first filing — under a single specification, so the coefficients can be read down a column. Magnitudes differ from the section-specific tables above.

Surprising language costs at registration and pays afterwards: atypical debut filings complete registration less often (−1.35pp per standard deviation), then survive the five-year proof of continued use more often (+1.35pp) and reach public markets more often (+0.04pp on a 0.56% base — the one specification in which a positive atypicality coefficient on listing survives, and it is a continuous term through a non-monotone profile, not the quintile contrast of Table 9). Having been early runs the other way: leading applications complete registration *more* often (+0.87pp) and then pass the five-year proof *less* often (−0.49pp on passing). Both survival rows point the same way at the year-ten renewal, four years later and on a sixth of the sample: atypicality +1.30pp and lead −0.63pp among marks that had already cleared the five-year proof. Whatever the five-year proof selects on, the renewal selects on it again rather than reversing it. The registration coefficient on lead is linear through a profile that is nearly flat (Section 3.1); the positive term reflects the asymmetry of the atypicality curve rather than a monotone gain from leading.

In the financing rows atypicality does nothing at one stage and a great deal at the next. Whether a debut owner ever raises a Regulation D round — a private securities placement, reported to the SEC on Form D — is unrelated to how atypical its filing was: −0.017pp per standard deviation among owners who completed registration, against a 2.61% base ($t = -0.8$). Among owners who did raise one, the same measure predicts *reaching* the public market: +0.49pp on a 3.49% base, about 14% relative ($t = 3.8$); the row is exploratory pending the Form D re-extract. Atypical language is invisible at the financing stage and informative afterwards, which is what an investor screen that does not read the text would look like — the text being free and public at the moment of filing. The comparison is among survivors, so it cannot separate that reading from selection on unobservables; and the null at entry is a null, not evidence of indifference. One construction matters: an exchange-registration marker on its own does not mean a firm listed *after* raising — 56% of matched owners carrying one registered before their first Regulation D notice, because established public companies also place securities privately — so the listing-after-round row is restricted to owners whose exchange registration postdates their first notice, which halves the coefficient and leaves it significant.

Three limits bound the financing rows. Regulation D notices are electronically filed only from 2009, so they rest on 2009–2018 debut cohorts. Owners are matched by normalized name, which succeeds for a small and unrepresentative minority favouring formally constituted entities, so every estimate is conditional on matchability. And the final row conditions on funding, realized after the filing being scored, so the coefficient is a contrast among survivors rather than a treatment effect: it establishes that information distinguishing eventual listers was already present in the text years earlier, not that the language caused the outcome.

Lead does not reverse anywhere after registration. It is negative at both use proofs, at financing with or without registration imposed, and at listing among the funded; the only row where it is not negative is SEC reporting among registered owners, where it is indistinguishable from zero (−0.004pp, SE 0.007). Leading filings fare worse at every stage where the market, rather than the examiner, is doing the selecting, and are never rewarded at any of them.

Why would having been early cost? The two axes are not independent in the obvious economic sense. What makes a filing leading is precisely that its industry moved toward its language, and an industry that has moved toward your language now describes its goods the way you describe yours. Being early therefore consumes the quantity that pays: the mark that was unusual in 2011 is ordinary by 2016 not because its owner changed anything but because the category arrived. On this reading atypicality is a position competitors erode, and lead measures how fast. The design cannot demonstrate that mechanism, since it cannot separate a category catching up with a filing from some third thing moving both. What can be said is weaker: a negative coefficient on lead at equal atypicality is what the mechanism would produce if it were real.

Read together, the record's stages sort the two properties cleanly. The examiner rewards middling conformity and cannot see lead at all. The market's first proof, five years on, punishes having been early — most where the language was moving fastest — and spares the unusual, though mostly the unusual that extends an established theme into a new class. The financing market reads none of it at entry. And the firms that came to hold most of the linked economy's value filed first descriptions in the words their industries already used. The pioneers of language are the ones with arrows in their backs; the settlers arrive writing prose everyone recognises, and they are few.

8 Robustness: what changes under other choices

Each result above is stated under one formulation chosen for legibility: one theme resolution, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 registration cohorts. This section reports what each of those choices costs, and in each case states the more complicated finding beside the simpler one the body uses.

8.1 Representation and weighting

A scoring is fixed by two choices. The *representation* is what the language is turned into: a distribution over 50 fitted themes, or a distribution over the class vocabulary's terms. The *weighting* is how days inside the reference window count: every day equal, or decaying by half every two years. Every reference is per-filing, anchored on the filing's own date (Section 2). Four builds exist:

Representation	Weighting	sd(lead)	sd(atypicality)
Themes	flat	0.102	0.698
Themes	half-life 2y	0.082	0.698
Terms	flat	0.155	1.101
Terms	half-life 2y	0.130	1.102

All four are compared on the 6.01 million filings every one of them scored. The first row is the production scoring.

Changing one choice at a time settles the ordering. Correlations of lead between builds differing in exactly one respect:

Choice varied	r (lead)	r (atypicality)
Weighting, flat vs half-life 2y (themes)	0.982	1.000
Weighting, flat vs half-life 2y (terms)	0.970	1.000
Resolution, 50 vs 200 themes[10]	0.636	0.828
Representation, themes vs terms	0.460	0.396

The weighting is close to irrelevant: whether recent language counts for more than distant language within the window barely moves the score, and it does not move atypicality at all to three decimals,

[10] The resolution row is computed on the five-year-proof sample rather than the 6.01 million-filing common sample of the other rows.

which is what the 45-degree rotation of Section 2 implies — the long axis is a property of the filing and the weighting acts almost entirely on the small residual. Resolution and representation matter most: quadrupling the number of themes leaves the two builds agreeing on 40% of the variance in lead, and themes against terms on less than a quarter. A robustness check that varies only the weighting is therefore close to a restatement; one that varies the representation or the resolution is a real test, which is why the term-scored replication in Appendix 8.11 and the sweep below are the ones the claims lean on.

8.2 Theme resolution

Fifty themes over a 62,168-term vocabulary is roughly one theme per industry; whether the findings depend on that coarseness is testable by refitting at higher T. Raising the theme count does not add coverage — the vocabulary is fixed, and a larger T re-partitions the same words — but it changes what counts as sitting far from an industry, and could change any sign. Re-fitting the model at four resolutions on the same stratified sample and the same analyzer, and re-scoring the software class under per-filing windows:

Themes	Proof contrast, lead	t	Proof contrast, atypicality	r with $T = 50$
50	+3.23pp	13.8	−14.01pp	—
100	+2.71pp	11.6	−13.69pp	0.690
200	+5.20pp	22.2	−13.45pp	0.640
500	+5.69pp	24.3	−14.49pp	0.641

Three readings, in descending order of how much weight they carry. The sign and the significance are untouched across a tenfold change in resolution, and so is atypicality, which stays within a point of −14pp throughout. The magnitude is not: it does not move monotonically in T, so no single resolution can be quoted as the software class's effect, and the figure this paper reports at $T = 50$ sits at the low end of the range rather than the high one. A fifth fit at $T = 1000$ exists and is excluded: its first run shipped undertrained models through an indentation bug, and the refit is not persisted. And agreement with $T = 50$ settles at 0.64 without decaying as T grows, which says the resolutions share a fixed core and differ in the remainder rather than drifting apart as the partition refines.

The corpus-wide contrast is steadier than the single class: at $T = 200$, scored for all 45 classes, it is +2.23pp against +2.29pp at $T = 50$. Only those two resolutions exist corpus-wide, so that is a two-point comparison, but it is the one the paper's estimates run on, and pooling evidently averages away much of the per-class resolution sensitivity.

8.3 Fitting noise at one resolution

The sweep leaves one question open. Scores at different resolutions agree at $r = 0.64$ to 0.69 on lead, and neighbouring resolutions agree no better than distant ones. That has two readings. The number of themes may genuinely change what is being measured. Or a theme-model fit may be unstable, with two runs settling in different local optima, in which case the sweep measures fitting noise and the resolution has little to do with it.

Holding T fixed and moving only the seed separates the two. A second fit at $T = 200$ on the same sample, the same vocabulary and the same number of passes, differing from the first in nothing but its random seed, agrees with it at $r = 0.79$ on lead and $r = 0.87$ on atypicality across 1.11 million scored software-class filings. The cross-resolution figures were 0.64 and 0.83. Most of what the sweep showed as resolution sensitivity is therefore present between two fits that differ only in the seed: for atypicality nearly all of it, for lead well over half. The resolutions are not measuring different things; each fit is a noisy draw on the same thing.

The estimate moves less than the scores do. On the identical sample, the five-year-proof lead contrast is +5.20pp under the first seed and +4.88pp under the second (SE 0.23 each), and the atypicality contrast −13.45 and −14.38pp. Two consequences follow. A single fit's lead score carries measurement error of roughly a fifth of its variance, which attenuates every coefficient on lead toward zero; correcting for it would enlarge the estimates, not shrink them. And the spread of magnitudes across the resolution table above, which does not move monotonically in T, should be read as a band about a third of a point wide from fitting noise alone rather than as a property of any particular T.

The substantive claim does not depend on any of this. The five-year-proof lead contrast on the 2002–2018 registrations each build scores runs +2.29, +1.95, +3.56 and +2.92pp across the four builds in the order tabulated above, and the atypicality contrast −4.39, −4.43, −3.63 and −3.73pp. Every sign holds; term scoring returns the larger lead contrast, so the headline estimate is the conservative representation.

8.4 Decayed reference windows

Inside the reference windows every day counts the same. The alternative weights recent language up, with a half-life of two years inside the same 1,826-day truncation, so that a term the class adopted last month is more familiar than one it adopted four years ago. Rescoring the corpus that way changes the scores little and the estimate somewhat. Flat and decayed lead correlate at 0.980 on the 3.10 million gate registrations and atypicality at 1.000 to three decimals. On the identical registrations, with quintiles cut within class and registration year, the top-minus-bottom gate contrast on lead is +2.29pp under the flat window and +1.95pp under the decayed one (SE 0.09 each); on atypicality it is −4.39 and −4.43pp. The decayed window therefore returns a lead penalty about a seventh smaller, with the same sign, the same precision, and the same profile. The flat window is kept in the body because it is the simpler statement; a reader who prefers recent language to count for more should take +1.95pp as the estimate.

8.5 Window width

The window is a parameter of the measure, not a property of it, and the five years used throughout was inherited from the class-year construction rather than chosen under the current one. Re-scoring the software class at several widths under per-filing references, and judging each by the five-year-proof quintile contrast it produces, gives +2.02pp at two years, +2.20 at three, +3.23 at five and +3.37 at seven, each on an identical sample of 452,911 registrations and each with a standard error of 0.23pp. Signal rises steeply from two to five years and is flat between five and seven: the +1.03pp gain from three to five is well outside the sampling error, the +0.14pp from five to seven is not. A ten-year window returns +2.89pp, but that figure is not comparable to the others — requiring both windows to lie inside the corpus costs 22% of the scored filings and 13% of the gate sample, dropping the earliest and latest registration years, which are also the years whose contrasts differ most. The apparent decline at ten cannot be separated from that change in composition and is not evidence about window width. On the range where the sample is fixed, seven years is nominally the peak and five is within noise of it while retaining more of the corpus, so five is defensible rather than optimal. The sweep is on one class and one outcome.

8.6 Mixing the two windows

"Leading" is relative to a reference, and the two references need not be the same width. A short past and a long future make lead a measure of foresight — whether the filing resembles where the class was heading over the next seven years more than what it wrote in the last three. A long past and a short future make it a measure of rupture — whether the filing has broken with what the class had established over seven years, judged against only the next three. Scoring every class on the production representation and per-filing windows at $W \in \{3, 5, 7\}$ years on each side, and evaluating all five variants on identical samples — the gate columns on the 3.10 million registrations 2002–2018, quintiles within class×registration-year; the registration columns on the scored filings 1995–2018, quintiles within class×filing-year:

Variant	Past, future (years)	First-gate failure Q5−Q1 (pp)	First-gate failure classes > 0	Registration Q5−Q1 (pp)	Registration Q3 − tails
Foresight	3, 7	+1.65 (0.09)	31 of 45	+1.09	+0.50
Symmetric	3, 3	+1.69 (0.09)	—	+0.79	+0.13
Symmetric	5, 5	+2.29 (0.09)	33 of 45	+0.71	+0.28
Symmetric	7, 7	+2.58 (0.09)	—	+0.48	+0.65
Past-rupture	7, 3	+3.00 (0.09)	37 of 45	+0.01	+0.33

The symmetric five-year row is the paper's FE-only estimate in Table 3, reproduced here to the decimal, which fixes the scale. The penalty rises monotonically along the mix, from +1.65pp under foresight to +3.00pp under rupture, and rupture is the more pervasive variant, positive in 37 of 44 classes against 31. In software the two are +1.82 and +5.07pp. So the use requirement punishes distance from the class's established language more than it punishes resemblance to its coming language; but foresight carries a penalty of its own, more than half the symmetric one, and the separation is a matter of degree rather than kind.

Registration is a different matter. On the signed axis the top-to-bottom contrast is at most a point on a 57% base and the interior-minus-tails depth at most two thirds of a point, under every mix. The examiner's response to lead, on this scoring, is close to nil.

8.7 Without conditioning on grant

Figure 13 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is nearly flat in L because two opposed gradients cancel: registration rises in lead (61.3% to 63.2%) while gate survival falls (44.2% to 40.8%); the caption carries the quintile figures. Conditioning on grant is what separates them. Panel B shows the conditional gate outcome on the same filings, and it is U-shaped rather than inverse-U (depth −2.11pp), with a clearly penalised leading tail consistent with the cleaner single-gate cohort of Figure 14.

8.8 Toggled filters, windows and parameters

Each new analysis in Sections 3–6 was rerun under its natural parameter toggles, on the identical data. Nothing reverses a headline claim; two auxiliaries move.

Variant	Baseline → variant	Reading
Gate window 3.5–9.0 / 4.5–8.0	lead +2.29 → +2.29 / +2.28pp	window immaterial
Registration cohorts 2002–09	+2.29 → +0.98pp	the era swing
Registration cohorts 2010–18	+2.29 → +3.11pp	penalty concentrated late
Atypicality, both cohort halves	−4.39 → −4.55 / −4.28pp	stable
Internet pattern, narrow core	goods web gap 0.0 → +2.0pp	divergence holds
Convergence arrival 1% / 4%	early→late +4.4 → +1.6	+5.9 → +1.8 / +3.8 → +2.3
Surge doubling 1.5× / 3×	in-surge net +1.3pp	+1.2 / +0.9pp
Wave entry window 4y / 8y	order flat	flat under both
Recombination floor 2 / 10	net −2.8pp	−1.2 / −3.6pp
Pair-lead presence 0.05	net +2.3pp	+3.1pp
Combination measures, class 035	009 values	recomb −3.8; pair-lead −1.4
Ladder, counsel / self debuts	reporting lead −0.21pp	−0.28 / −0.04pp
Ladder, 2009–13 / 2014–18	reporting atyp. +0.11pp	−0.01 / +0.20pp

Three of these change what a reader should carry. The lead penalty at the gate belongs to the 2010–2018 cohorts; the 2002–2009 half carries a third of it, which is Section 5.1's era swing seen from the other side. The ladder's gradients belong to counsel-represented debuts — among self-filers the reporting gradient is a null — and its atypicality gradient to the 2014–2018 half. And the pair-lead penalty, +2.3pp in the software class under every parameter tried there, is −1.4pp in advertising and retail, so it is reported as a bound for one class rather than a general fact. All artifacts are under `paper/v3/_eval/`.

8.9 Changing the time period

The registration cohorts the gate estimates use begin in 2002 because that is the conventional cut for this corpus, and end in 2018 because the cancellation record runs to April 2026. Section 5.1 carries the series back to the first scoreable cohorts: the penalty was two to three times larger for marks filed in the late 1990s than for any cohort since, reversed for filings made 2000–2004, and returned from 2008, with the swing inside the four technology classes and, within them, in which themes each era's leading filings were drawn from. Within the 2002–2018 window the penalty is zero to negative for the 2002–2005 cohorts (−2.3pp for 2003) and runs +1.8 to +5.4pp for every cohort from 2008 on the wide-window raw series. The 2019 cohort, evaluable only on the narrow 6.5–7.0 observation band for registrations issued early enough in that year (78,408), is consistent in sign at +5.6pp. Cohorts from 2020 have no observed proof. The pooled estimate is therefore an average over a penalty that was not constant, and a reader who wants the penalty for a period should read Figure 5 rather than the pooled row.

8.10 Lead is a small residual

Past-facing and future-facing KL are highly correlated however scored. On the complete class-009 corpus under the production scoring they correlate at $r = +0.988$ (n = 1,109,643). The lead L is therefore a small residual of two large, stable quantities: both levels have a standard deviation near 0.70 nats about a mean of 1.46, while L has a standard deviation of 0.109 — about a sixth of either level's spread. Only 1.2% of the two levels' combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels, and under term scoring the dominant such thing is document length. Term-scored atypicality correlates with the log of a filing's distinct term count at −0.681 in class 009 and −0.620 in class 035, for the reason set out in Section 2: the divergence charges a filing for the narrowness of its own distribution, and that narrowness is fixed by how many distinct terms it uses. Theme scoring weakens the coupling, taking the same correlations to -0.217 and -0.098. The consequence for the lead is direct: the standard deviation of L rises 2.2–2.5-fold across deciles of A under term scoring and 1.5–1.8-fold under theme scoring, so a unit of lead means roughly the same thing everywhere only in the theme representation. Appendix D shows this as a length surface over the two axes. Two qualifications. Theme scoring returns a finite value for about 65% of filings, so the two columns are not drawn from the same set; re-estimating the term-scored correlations on exactly the theme-scored subset barely moves them, which puts the contrast down to the representation rather than the sample. And "flat" means no monotone length gradient rather than constant: class 035 under theme scoring is mildly U-shaped in A. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.42 on the 6.01 million filings both score), which gives their agreement on the mark-level patterns evidential force and their disagreement on the firm-level patterns diagnostic force (Section 8.12). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT "most innovative" lists sit +0.09 standard deviations above the panel mean under term scoring ($p = 0.22$), +0.14 under theme scoring ($p = 0.07$), and +0.08 at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with L on the same panel (Pearson +0.010).

8.11 Term scoring as a cross-check

The snapshot cross-check on the single one-gate-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: -5.4pp survival across quintiles under theme scoring and -5.7pp under term scoring on the identical sample (Figure 14). Under term scoring, gate survival *rises* in the raw KL level (+6.5pp past-facing, +7.6pp future-facing): atypical text survives the use requirement better while leading text survives it worse — the same two-axis separation as Section 6.5, at the mark level.

8.12 Term scoring at the firm level

Under term-level scoring, leading filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in SEC EDGAR rises in term-L from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched firm-year panel, trailing 3-year mean term-L predicts gross margin at +2.2pp per standard deviation ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the published panel through 2024), with past-facing surprise entering positively and future-facing negatively; and firm-year term-L predicts excess stock returns of +1.5 to +5.0pp per standard deviation at one-to-four-year horizons. A larger debut-only return figure (+28pp at 4 years) fails a sample-size diagnostic and is not claimed.

None of the signed versions survives distribution scoring. On identical samples, the debut listing gradient in signed theme-L is flat to declining, and the margin coefficient reverses: -1.3pp per standard deviation ($t = -1.8$) at $T = 50$, -2.2pp ($t = -3.1$) at $T = 200$, and -0.8pp ($t = -1.1$) at $T = 500$, against the term-level +2.3pp; the past-/future-facing decomposition flips sign and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within length bands it is U-shaped rather than rising, with the monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions). The mark-level results are unaffected — they hold under both scorings at every resolution.

Vocabulary blindness does not rescue the signed signal. The theme model's vocabulary is sample-built, so rare and emergent terms are invisible to it; if the firm signal lived exactly there, theme scoring would erase a real signal rather than an artifact. Computing each filing's invisible-vocabulary share (63.8% of class-009 term *types* but only ~9% of token *mass*): the share itself is a strong firm marker (P(EDGAR) 2.9% to 5.5% across its quintiles; gate survival +5.7pp), but *within* invisible-share strata the signed L gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (−6.9 and −8.3pp). The term-level firm signal is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term-L mechanically. Theme count is a partition dial, not a coverage dial — a crypto/blockchain theme exists at $T = 200$ but not at $T = 50$ or $T = 500$, because raising T re-partitions the fixed vocabulary along dominant axes — so resolution sweeps test granularity, not coverage, and the invisible-share split is the test that binds.

The registration filter does not explain the divergence. Dropping the registration conditioning leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), and an examiner channel touching 4.3% of failed applications cannot generate several-point quintile gaps.

Term-resolved text-novelty measures can therefore manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic.

8.13 Functional form

Four shapes were fitted to each profile by weighted least squares on 40 equal-count bins (Figure 15): linear, quadratic, an exponential $a + be^{cx}$, and a free-vertex symmetric form $a + be^{c|x-x_0|}$ meant to catch a U directly. Selection is by AIC.

Outcome	Axis	R^2 line	R^2 curve	ΔAIC	Shape selected
Registration	atypicality	0.38	0.92	77.1	curve, peak at +1.04
Registration	lead	0.08	0.17	1.9	(quadratic; weak)
Use-proof	atypicality	0.74	0.74	0.0	linear
Use-proof	lead	0.44	0.55	4.3	V, vertex −1.50
SEC reporting	atypicality	0.74	0.84	15.2	curve, peak at +2.17
SEC reporting	lead	0.13	0.59	26.3	V, vertex +0.30
listed only	lead	0.19	0.46	12.2	V, vertex +0.41
reporting, not listed	lead	0.07	0.67	37.2	V, vertex +0.24

The winning form is a free-vertex $a + b\,e^{c|x-x_0|}$. On the atypicality rows it is a genuine curve, with fitted exponents between 0.5 and 1.2 — a hump that peaks about one standard deviation above mean atypicality at registration and about two above it at SEC reporting. On every lead row the exponent collapses to the order of 10^{-4}, which turns $e^{c|x-x_0|}$ into $1 + c|x - x_0|$: what wins there is a V, linear in distance from a vertex. A V beating a parabola says the turn is sharper than a quadratic allows, not that anything is multiplying.

Vertices are in standard deviations of the axis across filings.

The two rows for SEC reporting are the outcome Section 6.5 and Table 9 actually run on; the listed and reporting-only splits below them show the same shape on each half, which is why the split does not change the reading.

AIC here is computed on 40 binned means, not on the millions of underlying filings, so it ranks shapes rather than testing them. The 77 at registration and the 12 to 36 at the public-reporting gates are decisive; the 1.9 at registration on lead and the 4.3 at the five-year proof are weak, and those two rows are better read as "no worse than linear" than as curvature.

8.14 Listing against reporting

"Appears in SEC records" mixes two populations: companies whose shares trade, and entities that file with the Commission without ever listing — private issuers, funds, subsidiaries registering securities. Splitting the 19,889 matched owner strings on whether their CIK carries a ticker in the Commission's own company-ticker file gives 10,556 listed against 9,333 that file financial statements without one.

It does not change the finding. Among registered debut owners whose debut filing is scored, 0.310% later appear as listed companies and 0.285% as reporting non-listed ones, and both profiles are U-shaped in lead with vertices at +0.41 and +0.24 standard deviations. The listed profile is slightly flatter and noisier, which is what the smaller and more selected outcome predicts. Whatever the signed axis is picking up at this margin, it is not an artifact of counting non-operating registrants as commercial successes, and it is not specific to the act of listing either.[11]

8.15 Filings are not independent draws

Some of them share an exact position with another filing, and outcomes cluster within owner — the within-owner correlation is 0.38 for gate survival pooled and is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The regression tables are unaffected, because the gate regressions cluster on normalized owner and the firm-level specifications carry one observation per owner. The raw quintile contrasts of Sections 4–5 — the era, theme, internet and surge tables — are not: their two-sample standard errors ignore the owner clustering, and their t-statistics overstate precision by roughly a factor of two, which the text of those sections now carries where the margins are close. Binning filings without the correction likewise overstates how much structure the corpus contains: on a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the gate outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason, and the near-zero pooled correlation between A and L reported in Section 2 does not survive cell by cell within class, so per-cell contrasts should not be read as within-class effects.

8.16 What remains unresolved

What drives the episodic pattern is not settled. Cohort and gate year move together, so post-crisis bet composition, app-era speculative filings, and calendar-time enforcement changes cannot be separated on this design. The representation is also fit on the pooled corpus; a vintage refit would close the remaining look-ahead channel.

A Corpus, scoring, and burn-in

A.1 Corpus and scoring

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work uses the complete

[11] Two limits on the split. The ticker file is a current snapshot, so a company that listed in 2003 and was acquired in 2011 appears in the non-listed tier; the tiers are therefore "currently listed" rather than "ever listed," which will misclassify older successes downward. And the base rates here are computed on the appendix's own panel, which conditions on the debut filing carrying a score and so is smaller than the body's sample.

software/electronics class (Nice 009; 1.81M filings); the diffusion application uses four classes totalling 2.32 million granted filings 1990–2024; the gate analyses use all 45 classes. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees by normalized-name join.

The theme representation is a $T = 50$ LDA fit on a stratified sample of 448,437 filings across all 45 classes (vocabulary of 62,168 unigrams and bigrams); every filing 1990–2024 is transformed to a dense 50-theme distribution, and past-facing and future-facing surprise are computed against per-filing theme references spanning 1,826 days on each side of the filing date, scoring 1995–2019. A filing whose window holds fewer than 500 same-class filings on either side is left unscored, since a thin reference inflates the divergence mechanically; that floor, together with the corpus edges, is why roughly two-thirds of filings carry a score. A term-level variant is retained alongside, on the same anchoring: token distributions (within-class document frequency ≥ 50) against per-filing windows of the same 1,826 days, Dirichlet-smoothed ($\alpha = 1$) over the class vocabulary. Because a goods/services description runs to a few dozen words, the divergence is needed only at the terms a filing actually uses, so the reference is carried as a running count vector and read sparsely. The term variant is used where vocabulary identity is the object (phrase transit) and as the comparison scoring throughout. Both representations are also computed under an exponentially decayed reference with a two-year half-life, truncated at the same window; Appendix 8.11 reports what changes.

A.2 Scores are interpretable from 1995

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

The signal depends on having enough past and future filings to populate the references; the 5-year past window is incomplete before 1990 and the future window after 2020, and both zones are excluded (Figure 16). The burn-in cut is set empirically: thin past references inflate past-facing surprise, and profiling that bias across all 44 sufficient-volume classes gives a median absolute bias of $\sim$2.0 nats in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any interpreted effect size. The 1995–1997 deviations ($\sim$0.15–0.17) are not burn-in, because mechanical contamination declines monotonically as the windows fill and cannot rebound. They reflect a genuine vocabulary shift in the corpus during the dot-com period. Per-class burn-in lengths estimated under a stability-band criterion track each class's noise floor rather than its reference density, so a single standard cut at 1995 is used, conservative by about two years.

B Examination rewards a middling specificity

Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure 18, panel C). The same shape appears on the past-facing level alone, more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure 17A).

The signed axis behaves differently. Completion on L falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing: $|L|$ correlates with atypicality at +0.31, so filings near zero lead are disproportionately low-atypicality filings, which complete poorly for the reason above. Split by atypicality, the trough is absent in the upper three fifths and survives in the lowest. The signed axis still enters a linear specification strongly (Table 10), but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing's language sits from its category, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most category-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

Refiling after abandonment is itself mildly related to vocabulary position, though not on the axis that matters here. Refiling rates are flat in atypicality (a middle-minus-tails difference of −0.06pp on a 3.6% base) but rise with lead, from 2.6% among the most lagging abandoned applications to 3.4% among the most leading.

C How refiled descriptions are rewritten

Figure 20 shows the goods/services description before and after a rewrite, for the 31,084 refilings in Section 3.3 whose text changed. A pair is an abandoned application and the same owner's later registration of the identical mark in the same Nice class within six years; the earliest registered refiling is kept. Counsel of record is read from the prosecution history of each filing separately, so a pair can change representation between attempts. Atypicality is on the production scoring; length is the word count of the normalized description.

The means understate the convergence, because it runs in both directions. By fifth of the original's atypicality, the change is +0.16, +0.14, +0.06, −0.06 and −0.39 nats from the most conventional fifth to the most atypical, and within a point of that profile in each of the four representation groups. Identical-text refilings change by +0.002 to +0.008 across the same fifths. On lead, the changed-text profile runs +0.046 to −0.072 and the identical-text profile +0.014 to −0.054; the difference is the rewriting, the rest is the reference windows moving with the later filing date.

D Term scoring measures length; theme scoring does not

Figure 21 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 200 terms — to near-white at the far end, where the median is eight. That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under theme scoring the same surface is a narrow band of roughly uniform tone.

Two filings are marked in both columns. Under the production scoring they behave differently from one another, which is the point. AMAZON.COM (1997) is at the 98th percentile of lead on terms and the 94th on themes, but only the 40th and 47th percentiles of atypicality, so both agree that it was early and neither mistakes it for strange. Apple's IPOD (2001) divides them: the 31st percentile of lead on terms against the 62nd on themes. The surfaces plotted here are on the class-year build the figure was generated from, where the length gradient is starkest; the percentiles just quoted are on the per-filing build used for every estimate.

Theme scoring is independent of document length, which term scoring is not; term scoring resolves novel word *combinations*, which theme scoring does not. Hence theme scoring for every estimate, since the length confound contaminates firm-level comparisons, and term scoring for every worked example, since the theme basis has no coordinate for the language the examples turn on.

E How three filings encode

Table 11 traces three filings through term scoring and theme scoring at $T \in \{50, 200, 500\}$ — three kinds of novelty, and how each representation handles them.

Amazon.com (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention ("computerized line," "line search," "search ordering"; class document frequencies 379, 134, 355). The filing's text reads "on line" as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword "on" and the bigram "line search" survives stopword removal. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under theme scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

Kombucha (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the theme vocabulary but, with no kombucha-like theme, the filing loads on a generic beverages theme at every T (Table 11).

Ethereum (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services themes at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain theme; there, the eleven occurrences of *software* dominate the theme assignment.

The pattern across the row is the measurement point in miniature: theme scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting theme exists. That the mark-level outcome results are nearly identical under both scorings despite this blindness is what licenses theme scoring as a robustness instrument; that the firm-level results diverge is what exposes them as artifacts of the lexical specificity term scoring additionally captures (Section 8.12).

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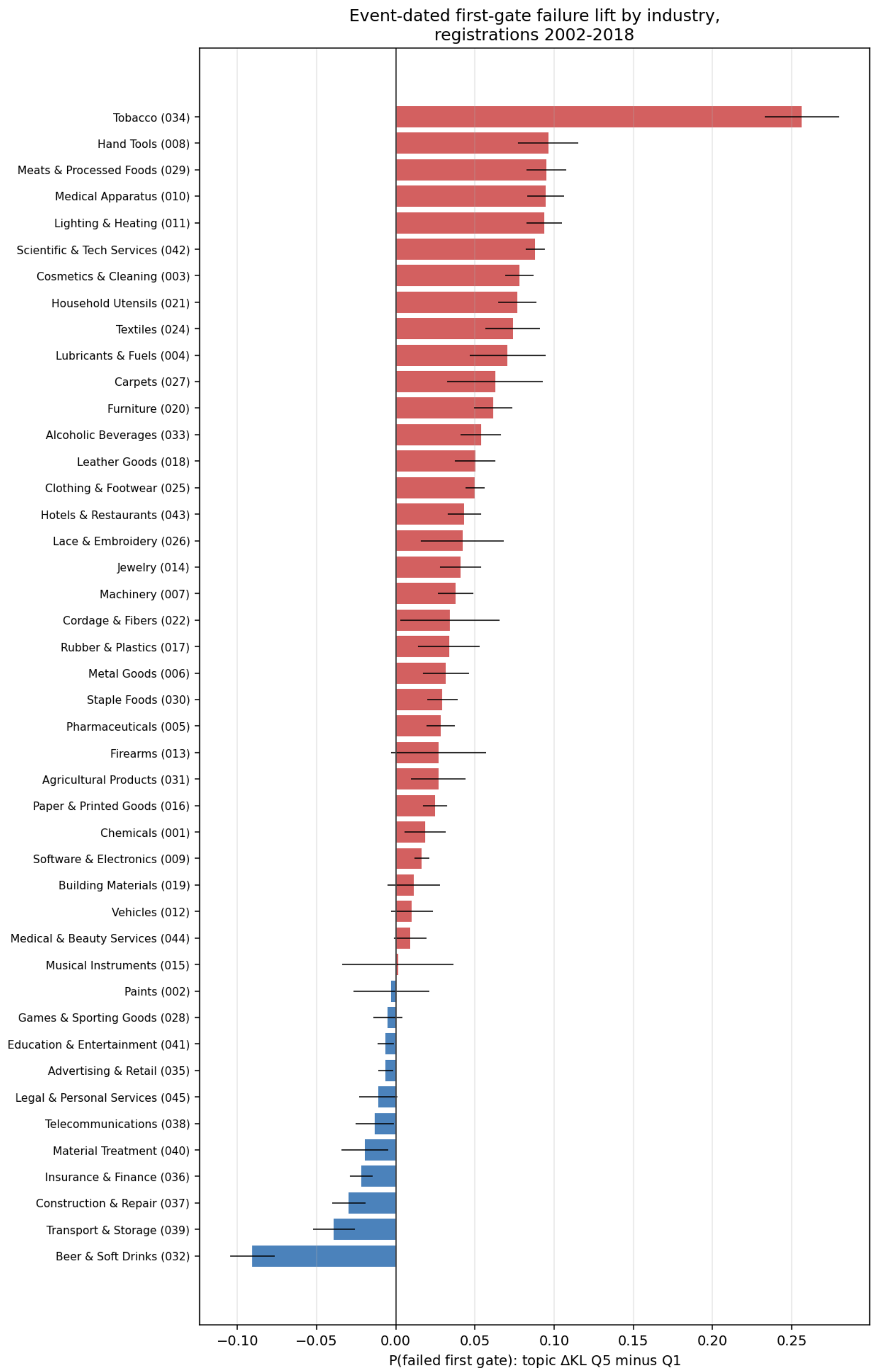

Figure 3: Failure lift at the five-year proof of continued use (lead top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts; classes need 5,000 scored registrations to enter. Positive bars indicate the leading penalty. Whiskers are two-sample binomial intervals, uncorrected for the within-owner correlation (0.38) and so narrower than clustered intervals; read Table 3 for precision. Per-class breakouts are in the online appendix.

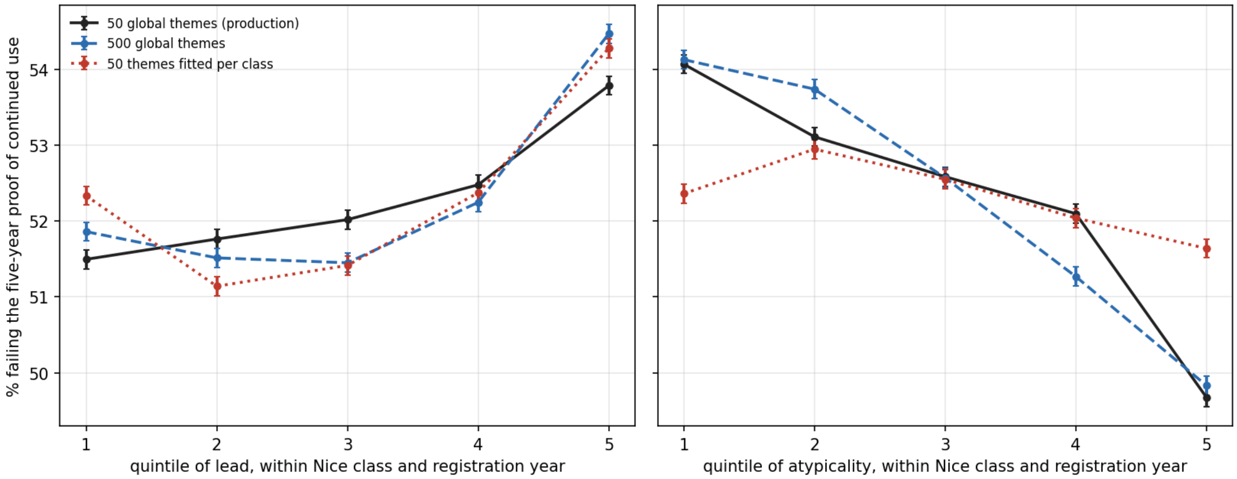

Figure 4: Failure at the five-year proof of continued use by quintile of lead (left) and of atypicality (right) under the three theme scorings, on the identical 3.10 million registrations 2002–2018; quintiles cut within Nice class and registration year, no controls. Whiskers are 95% binomial intervals per quintile, uncorrected for owner clustering. The lead gradient survives every scoring; the atypicality gradient does not survive per-class themes.

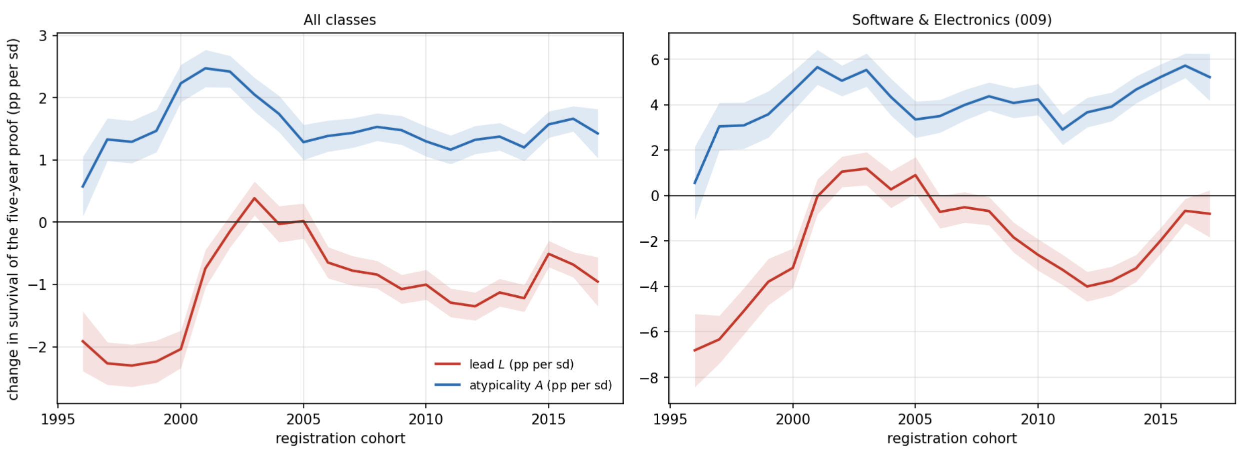

Figure 5: Survival of the five-year proof of continued use, per registration cohort, as continuous coefficients: the change in survival, in percentage points, for a one-standard-deviation increase in lead (red) and in atypicality (blue), entered jointly, both standardized within Nice class and cohort. Left: all classes pooled ($n = 3{,}150{,}442$ settled registrations); right: software and electronics alone. Bands are 95% intervals on OLS standard errors, uncorrected for the within-owner correlation of outcomes (0.38), so they understate the uncertainty somewhat.

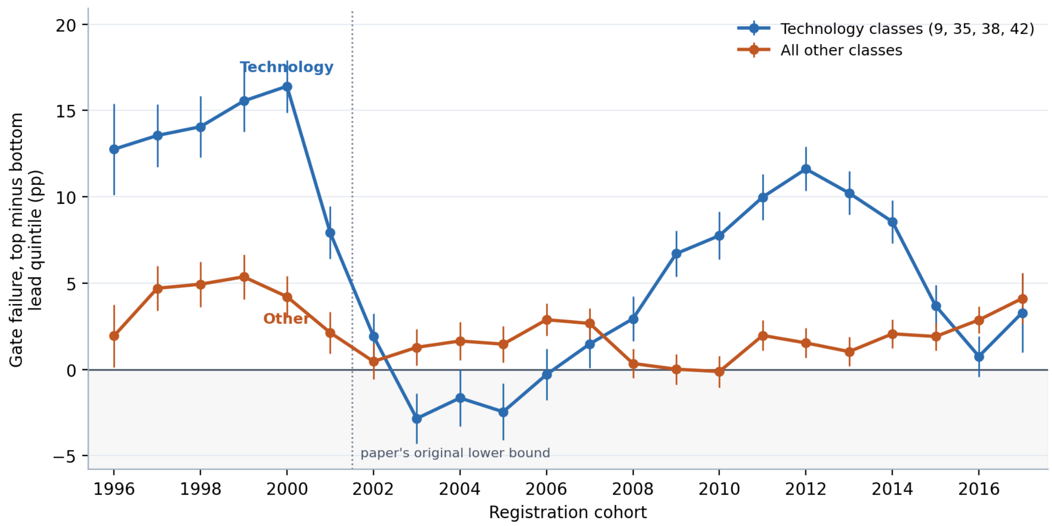

Figure 6: The leading penalty at the five-year proof by registration cohort, technology classes (009, 035, 038, 042) against all others. Each point is the failure rate of the most leading fifth of that cohort's registrations minus that of the most lagging fifth, lead quintiles cut within cohort and group, no controls, with 95% intervals, on settled registrations. Outside technology the penalty is small and almost flat across two decades; inside technology it swings from +16pp for the 2000 cohort to −3pp for 2003 and back to +12pp for 2012. The dotted line marks the 2002 cut that the cohort samples of Section 4 use.

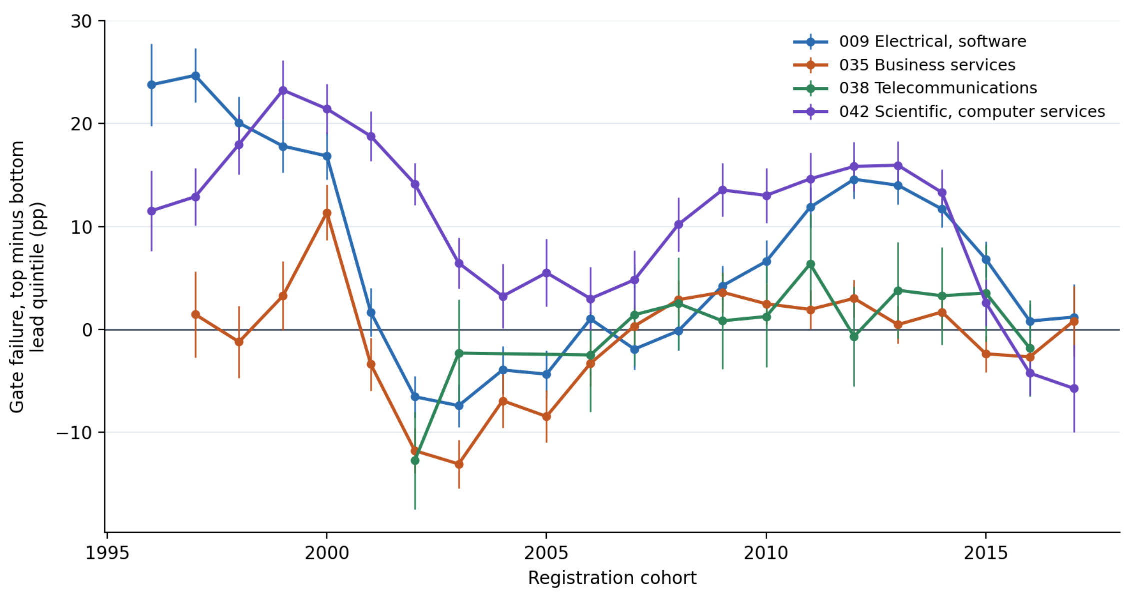

Figure 7: The leading penalty at the five-year proof by registration cohort, one line per technology class. Each point is the failure rate of the most leading fifth of that class and cohort's registrations minus that of the most lagging fifth — lead quintiles cut within class and cohort, no controls — with 95% intervals, on settled registrations. Three classes cross zero between 2001 and 2006; scientific and computer services does not.

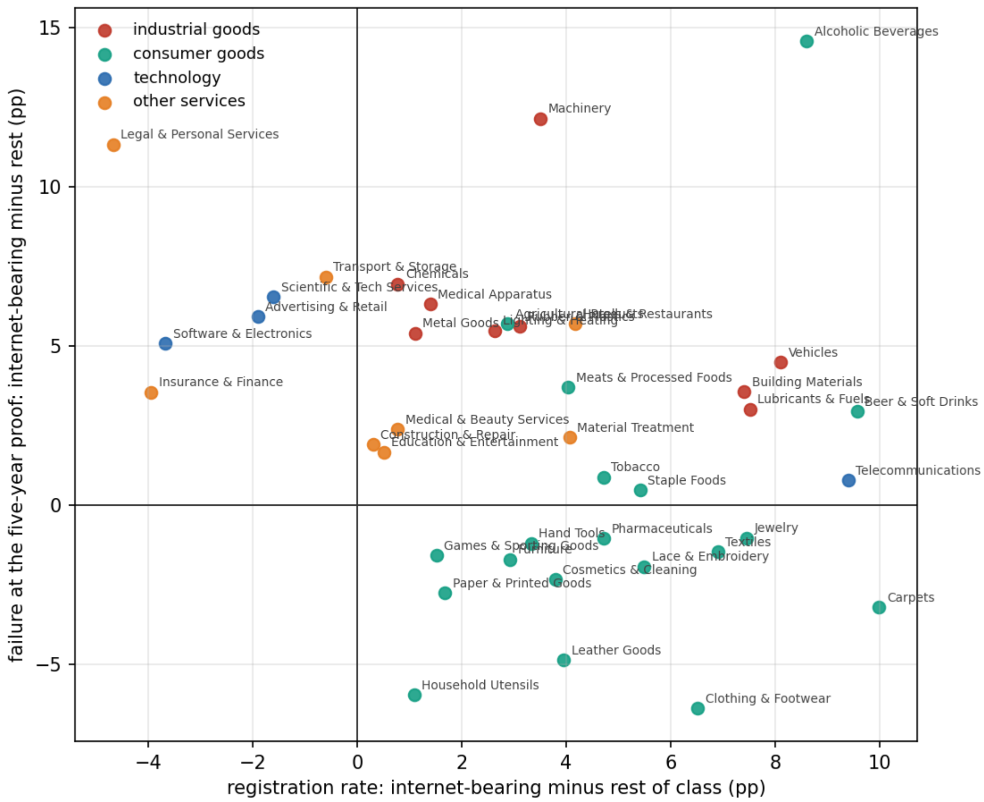

Figure 8: Each Nice class's internet-bearing filings against the rest of the class, on both steps at once. Horizontal: registration rate of internet-bearing class-records minus the rest of the class, filings 1995–2018. Vertical: failure at the five-year proof of continued use, internet-bearing minus the rest, registrations 2002–2018. Classes with fewer than 500 internet-bearing registrations are dropped. Classes in the upper-right quadrant grant internet filings more often and lose them more often.

Table 7: Internet-bearing filings against the rest of their Nice class. Registration is the share of class-records filed 1995–2018 that reached registration; failure is the share of unique registrations 2002–2018 cancelled for non-use at age 4.0–8.5 years. Classes with fewer than 500 internet-bearing registrations are omitted.

Class		Web share (%)	Registration (%) web	Registration (%) rest	Gate failure (%) web	Gate failure (%) rest
001	Chemicals	1.8	65.0	64.3	46.1	39.1
003	Cosmetics & Cleaning	2.8	58.0	54.2	52.1	54.4
004	Lubricants & Fuels	4.6	63.8	56.3	50.9	47.9
005	Pharmaceuticals	1.9	53.3	48.6	52.4	53.5
006	Metal Goods	3.5	68.1	67.0	44.5	39.1
007	Machinery	2.8	73.3	69.7	50.9	38.7
008	Hand Tools	3.6	65.8	62.5	42.4	43.6
009	Software & Electronics	18.7	53.3	57.0	56.5	51.4
010	Medical Apparatus	3.0	59.8	58.4	51.7	45.4
011	Lighting & Heating	3.0	67.4	64.8	51.4	45.9
012	Vehicles	3.4	69.0	60.9	49.6	45.1
014	Jewelry	7.4	63.2	55.8	54.7	55.7
016	Paper & Printed Goods	15.1	57.8	56.1	47.9	50.7
017	Rubber & Plastics	2.2	70.7	67.6	44.1	38.5
018	Leather Goods	8.0	61.4	57.4	50.5	55.4
019	Building Materials	2.0	71.4	64.0	47.5	43.9
020	Furniture	4.2	62.2	59.2	47.0	48.7
021	Household Utensils	5.1	60.1	59.0	44.5	50.5
024	Textiles	6.0	62.8	55.9	49.7	51.2
025	Clothing & Footwear	5.6	56.1	49.6	52.4	58.7
026	Lace & Embroidery	7.6	61.2	55.7	52.0	53.9
027	Carpets	5.6	66.8	56.8	43.7	46.9
028	Games & Sporting Goods	6.3	55.1	53.6	54.1	55.7
029	Meats & Processed Foods	2.5	59.7	55.7	53.3	49.6
030	Staple Foods	2.6	61.4	56.0	52.3	51.8
031	Agricultural Products	2.5	60.9	58.0	49.5	43.8
032	Beer & Soft Drinks	2.4	58.7	49.1	56.2	53.3
033	Alcoholic Beverages	1.2	61.3	52.7	56.9	42.3
034	Tobacco	2.9	53.8	49.1	55.0	54.2
035	Advertising & Retail	40.7	59.3	61.2	57.9	52.0
036	Insurance & Finance	18.9	58.6	62.6	52.4	48.9
037	Construction & Repair	10.4	67.8	67.5	49.0	47.1
038	Telecommunications	57.6	54.5	45.1	60.8	60.1
039	Transport & Storage	21.3	62.5	63.1	54.8	47.6
040	Material Treatment	9.6	68.0	63.9	49.4	47.3
041	Education & Entertainment	35.0	61.1	60.6	54.2	52.5
042	Scientific & Tech Services	39.3	57.8	59.4	58.0	51.4
043	Hotels & Restaurants	9.1	62.9	58.7	53.8	48.1
044	Medical & Beauty Services	20.6	62.9	62.1	53.7	51.3
045	Legal & Personal Services	41.5	60.3	65.0	60.1	48.7

Figure 9: The internet gap in failure at the five-year proof, inside each class. *Left:* internet-bearing minus other registrations' failure rate, by registration cohort in years since the internet pattern reached 2% of the class's filings, averaged across classes (band: one standard deviation across classes). *Right:* five classes in calendar time. Cells with fewer than 100 registrations in either group are dropped.

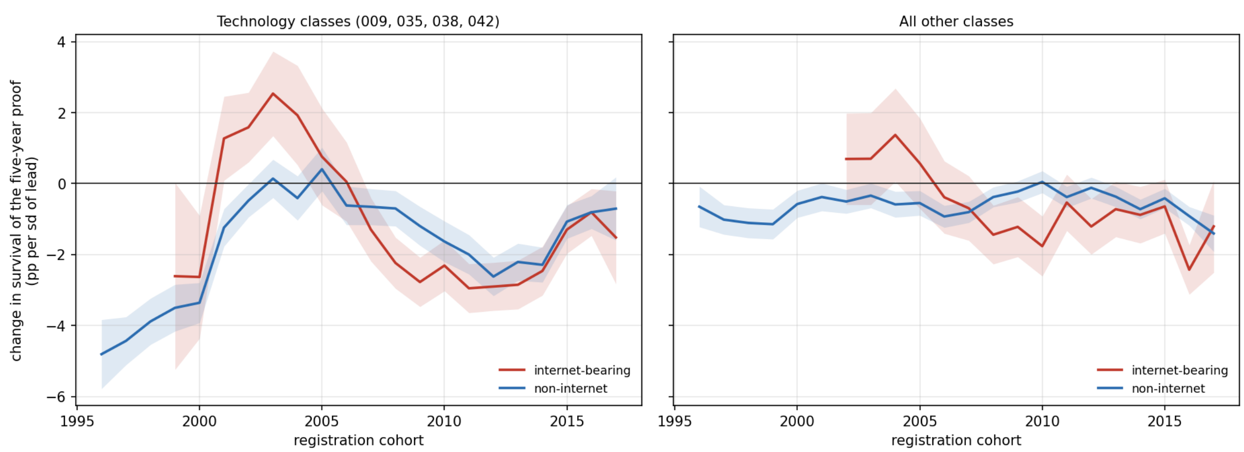

Figure 10: The per-cohort lead coefficient of Figure 5, split by the curated internet text pattern: change in survival of the five-year proof per standard deviation of lead, atypicality held, standardized within Nice class and cohort. Left: the four technology classes; right: all other classes. Bands are 95% intervals on OLS standard errors; cohort cells under 2,000 registrations are dropped, which is why the internet-bearing lines start late.

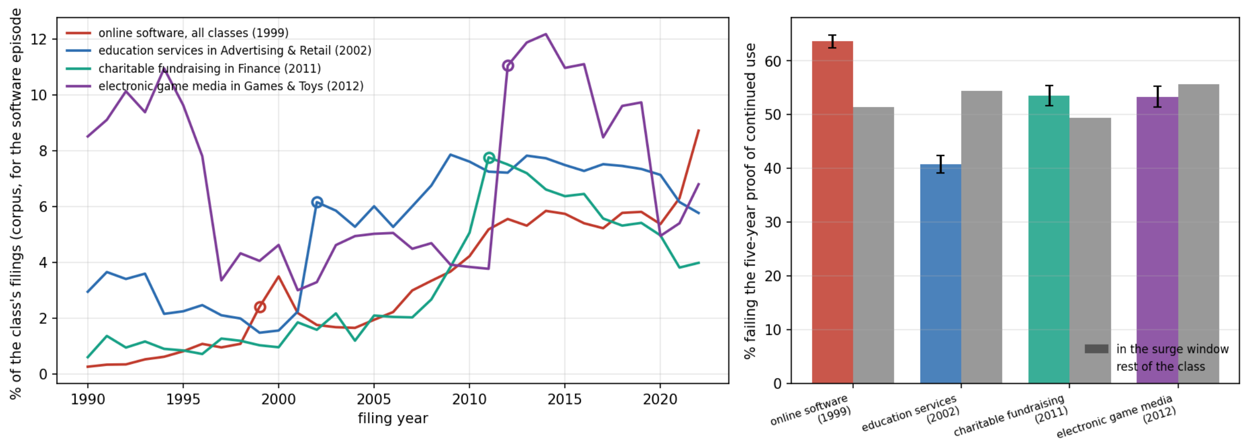

Figure 11: Four surge episodes under the 500-theme model. *Left:* the theme's share of its class's filings by filing year (for the online software episode, of all filings corpus-wide), on the 25% systematic sample the surge screen uses; the open circle marks the surge year, the first in which the share doubled its own three-year prior mean above the threshold. *Right:* failure at the five-year proof of continued use for registrations whose dominant theme was the surging one, filed in the three-year surge window (colored, 95% intervals), against the rest of the same class in the same cohorts (grey).

Table 8: The ladder by fifth of debut lead and of debut atypicality. Rates in percent of owners whose first registered filing was made 2009–2018 ($n = 999{,}442$); fifths cut within Nice class, debut year, and fifth of description length. The last column is the most-minus-least contrast with its t-statistic.

Rung	Axis	Q1	Q2	Q3	Q4	Q5	Q5−Q1 (pp; t)
Funded	lead	1.86	1.73	1.61	1.57	1.80	−0.06 (−1.4)
	atypicality	1.68	1.70	1.79	1.76	1.64	−0.04 (−1.1)
Reporting	lead	0.59	0.49	0.40	0.34	0.38	−0.21 (−9.4)
	atypicality	0.42	0.36	0.43	0.46	0.53	+0.11 (+5.2)
IPO	lead	0.24	0.21	0.14	0.13	0.17	−0.07 (−4.5)
	atypicality	0.15	0.15	0.18	0.21	0.22	+0.07 (+4.8)

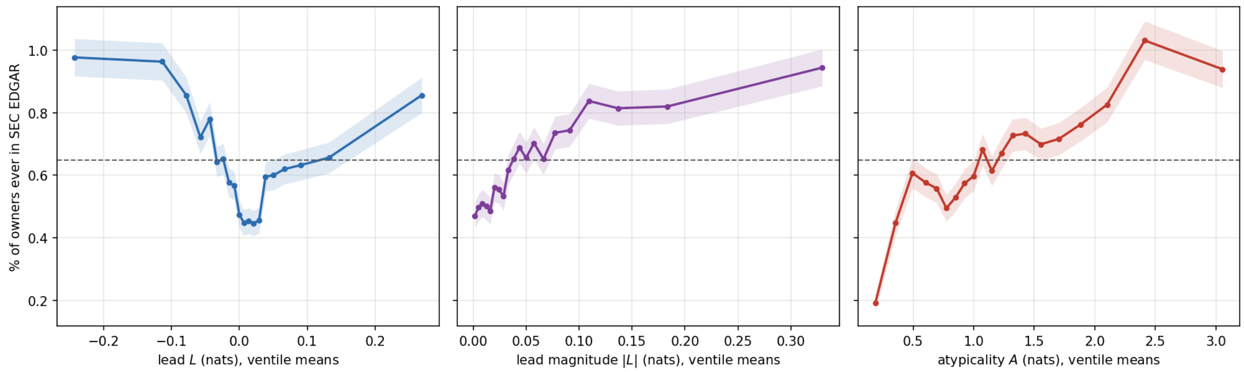

Figure 12: Share of registered debut owners (first registered filing 1995–2018, n = 2,066,600) ever appearing in SEC EDGAR, in twenty equal-count bins of debut lead L, lead magnitude $|L|$, and atypicality A, cut on the pooled distribution. Bands are 95% binomial intervals; the dashed line is the 0.65% base rate. These are raw rates carrying class, year and length composition; Table 9 gives the design-based estimates, under which the atypicality gradient is absorbed and the signed-lead U remains.

Table 9: Debut listing and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,968 registered debut filings 1995–2018, one observation per owner, class×filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class×year. Base rate 0.481%.

	Coefficient (pp)	SE
Atypicality quintiles, $A = \frac{1}{2}(K^- + K^+)$ *(Q1 omitted)*		
Q5, FE only	+0.017	0.018
Q5, FE + log length	+0.000	0.018
Q5, FE + log length, on K^- alone	−0.003	0.018
Signed lead quintiles, FE + log length + KL levels		
Q3	−0.133	0.020
Q4	−0.187	0.021
Q5	−0.158	0.024
Decomposition, FE + log length		
lead magnitude $|L|$ (pp per sd)	+0.067	0.009
signed lead L (pp per sd)	−0.047	0.006
log description length	+0.056	0.004

Standard errors in place of significance stars; see the note to Table 3.

Table 10: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner's first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the average of the two KL levels; *lead* is the signed difference. Cohort windows differ because each maintenance gate needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition. The first gate is a dated cancellation for non-use; the second is whether a registration that cleared the first was renewed rather than cancelled or expired, restricted to 2002–2013 cohorts, since a year-six and a year-ten death share a terminal status and are separable only this way. This table alone is estimated at $T = 200$ topics rather than the $T = 50$ used elsewhere; the sign and ordering of every row are unchanged at $T = 50$.

			Unsigned		Signed
Outcome *(population)*	n	Base	Atyp.	$\lvert\Delta\mathrm{KL}\rvert$	Lead
Reached registration *(filed)*	3,161,552	56.11%	−1.352 (0.033)	−0.307 (0.044)	+0.872 (0.028)
Passed gate 1, yr 6 *(registered)*	1,260,392	41.82%	+1.349 (0.054)	−1.150 (0.082)	−0.487 (0.052)
Passed gate 2, yr 10 *(gate 1)*	297,944	55.92%	+1.296 (0.108)	−0.297 (0.162)	−0.629 (0.105)
Raised Reg D round *(filed)*	1,535,004	2.53%	+0.003 (0.015)	+0.148 (0.027)	−0.149 (0.016)
Raised Reg D round *(registered)*	916,448	2.61%	−0.017 (0.020)	+0.115 (0.035)	−0.114 (0.021)
Listed after round *(funded)*	38,886	3.49%	+0.493 (0.129)	+0.608 (0.170)	−0.513 (0.104)
Became SEC-reporting *(registered)*	1,773,938	0.56%	+0.042 (0.007)	+0.040 (0.011)	−0.004 (0.007)

Standard errors in parentheses, in place of significance stars; see the note to Table 3.

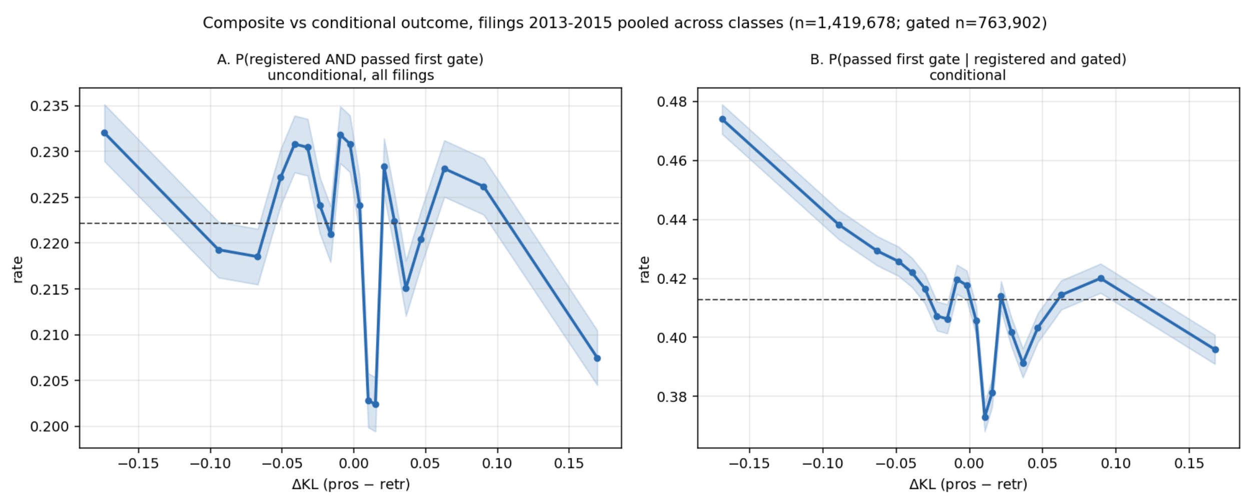

Figure 13: Filings 2013–2015 pooled across classes, theme-scored on per-filing reference windows (n = 1,419,678; gated subset n = 763,902). *A:* the unconditional composite P(registered and passed the five-year proof) over all filings. *B:* P(passed | registered and gated) on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

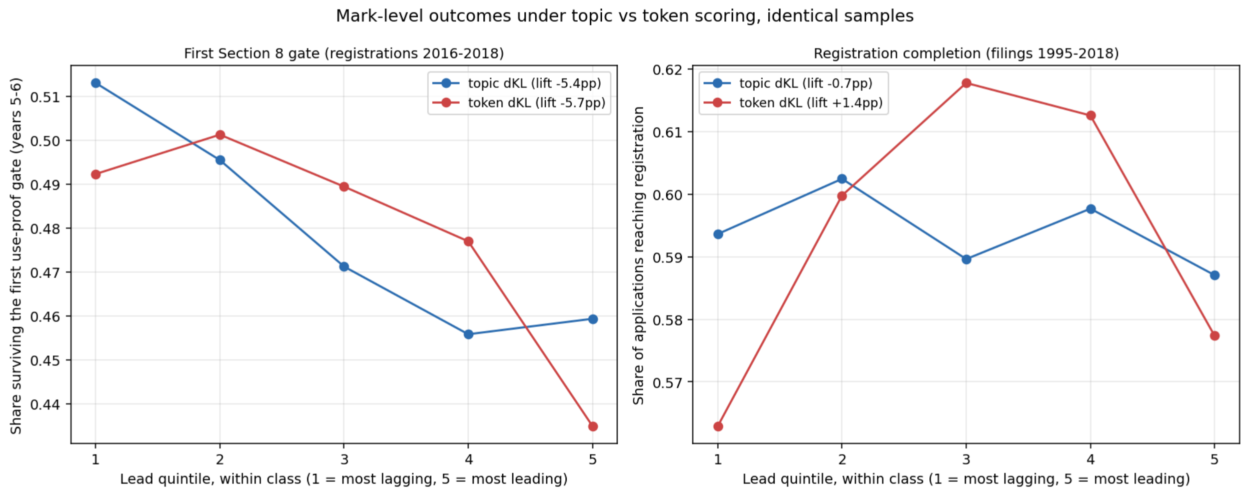

Figure 14: The two mark-level outcomes under theme and term scoring on identical samples, pooled across classes, by within-class lead quintile (1 most lagging, 5 most leading). *Left:* the share of registrations 2016–2018 that survived the five-year proof of continued use at years five to six, read from terminal status codes, where the two representations agree closely. *Right:* registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth +0.048) and absent under theme scoring (depth −0.001). The gate result is scoring-robust; the registration curvature on this axis is not, and Section 3 reports only the unsigned version.

Table 11: Top theme loading for three filings, by representation. Term scoring resolves the novel content of all three; theme scoring resolves none of it at any tested resolution.

	AMAZON.COM (1997)	KOMBUCHA (1997)	ETHEREUM (2018)
novel content (term)	line search, search ordering	kombucha, saccharose	blockchains, distributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, development (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

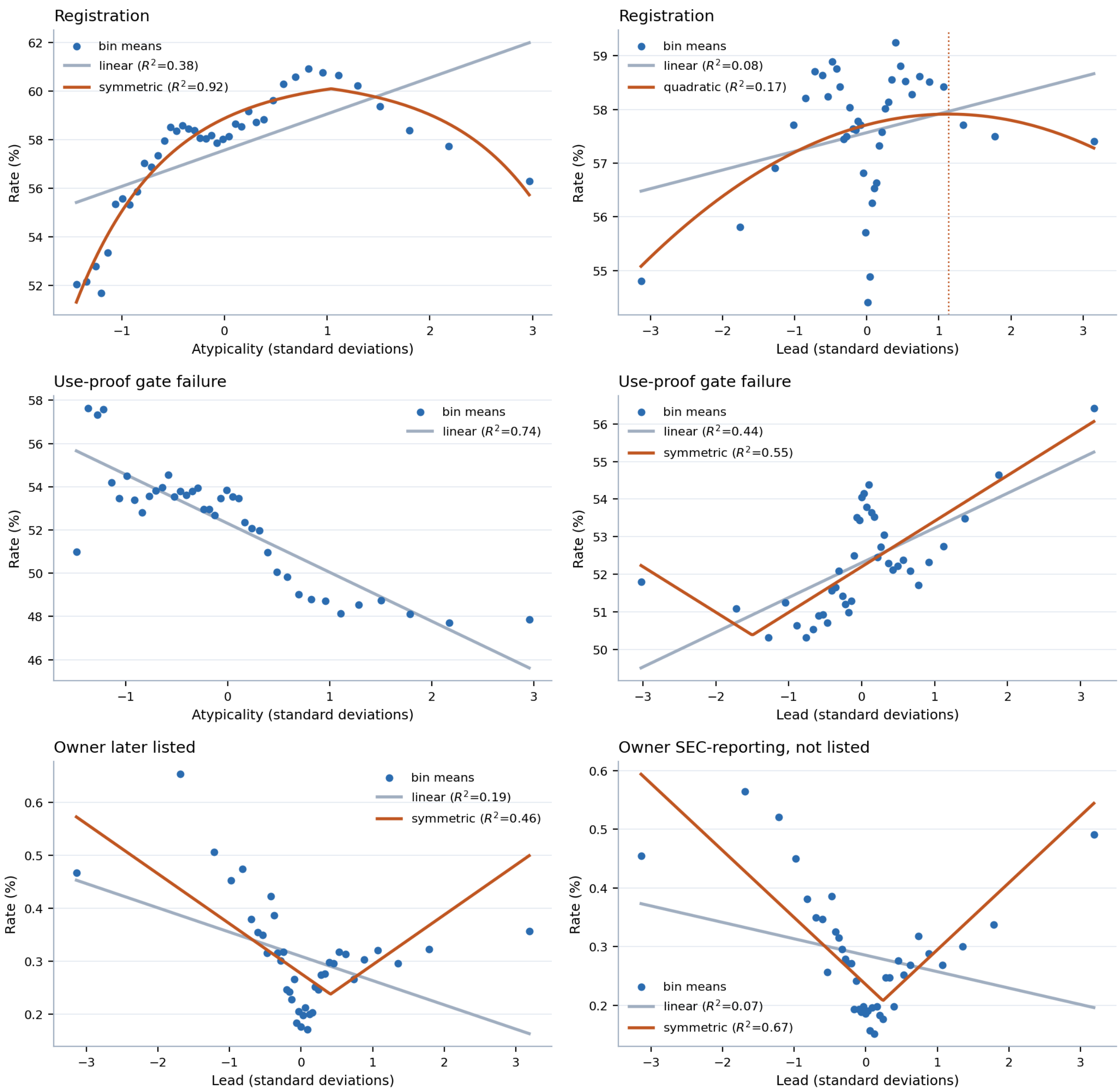

Figure 15: Binned profiles at the three gates, with the linear reading the paper's contrasts assume and the shape selected by AIC where that is not linear. Forty equal-count bins; the dotted line marks the fitted vertex. The bottom row is the case that matters: a linear fit through the public-reporting profile slopes downward while the data turn up at both tails.

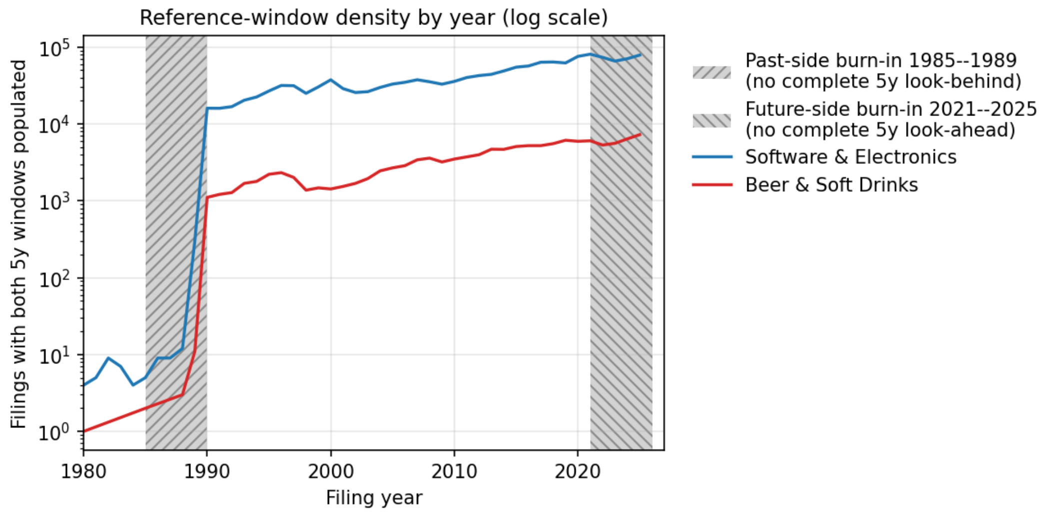

Figure 16: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete. The dotted line marks the 1995 analysis cut. The windows decide the score at the filing date; the gate outcome is read separately, at registration age six to seven.

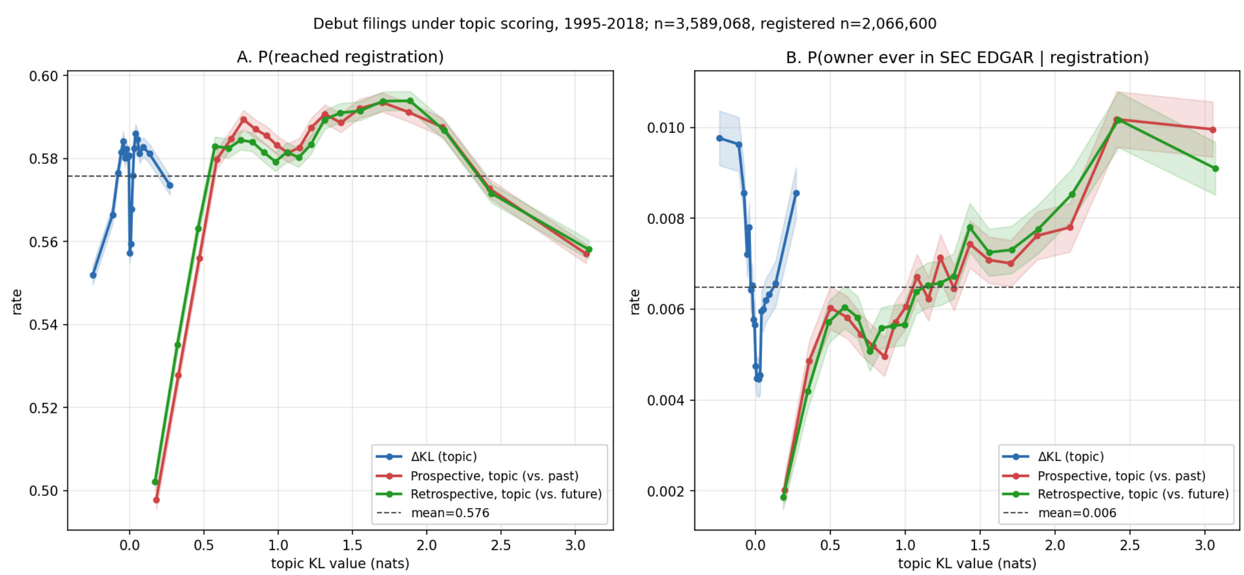

Figure 17: Outcomes for debut filings (owner's first-ever filing) across all 45 Nice classes under theme scoring, filing years 1995–2018, in quantile bins with 95% intervals, quantiles cut on the pooled distribution. $n = 3{,}589{,}068$; registered subset $n = 2{,}066{,}600$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR, the Commission's public filing system | registration).

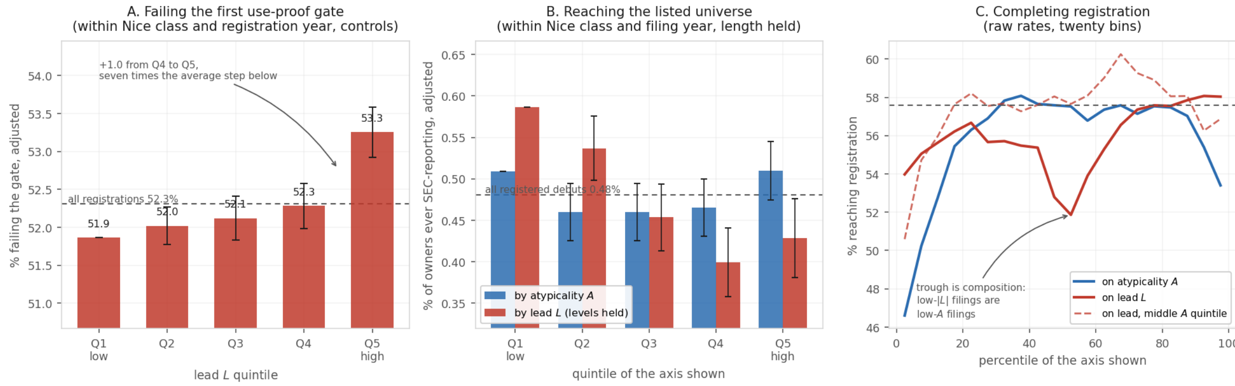

Figure 18: Quintile profiles for the three outcomes. *A:* the share of registrations failing the five-year proof of continued use, by lead quintile, adjusted within Nice class and registration year with the drafting and filer controls of Table 3; the dashed line is the rate for all registrations. The profile rises gently and monotonically through the fourth quintile and then steps up: about seven tenths of the top-to-bottom gap is the single step from the fourth quintile to the fifth, seven times the average step below it. *B:* the share of first-time filers whose owner ever reaches SEC reporting, by quintile of atypicality and of lead, adjusted within Nice class and filing year with description length held. *C:* raw registration rates for debut filings in twenty bins, cut on the pooled distribution — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate.

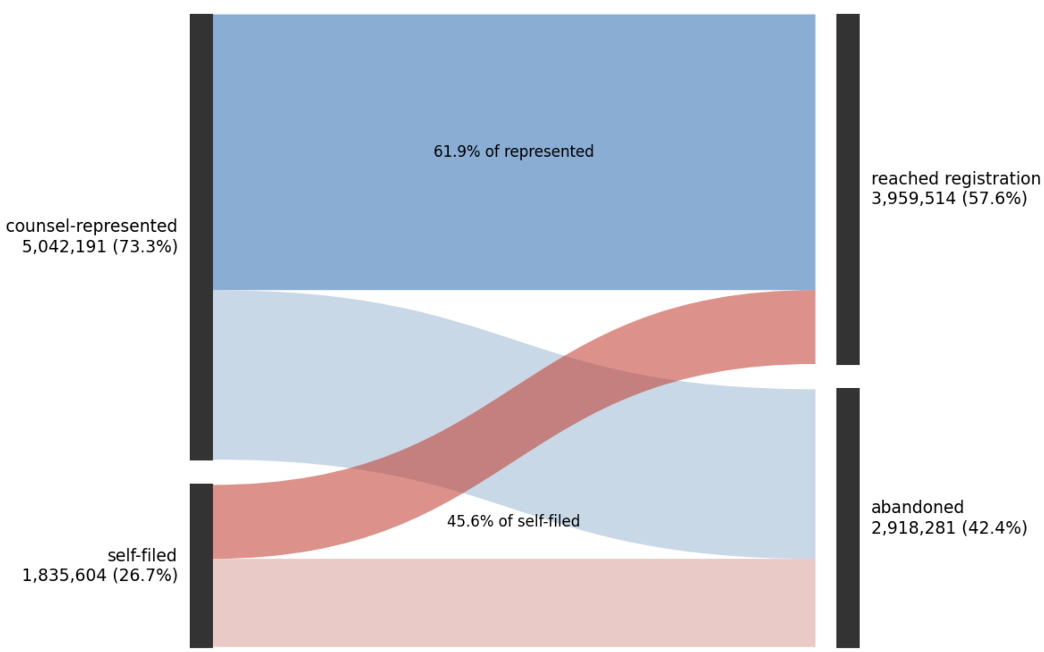

Figure 19: The registration step as a flow, unique serials filed 1995–2018 ($n = 6{,}877{,}795$), split by counsel of record on the filing. Represented applications reach registration at 61.9%, self-filed at 45.6%.

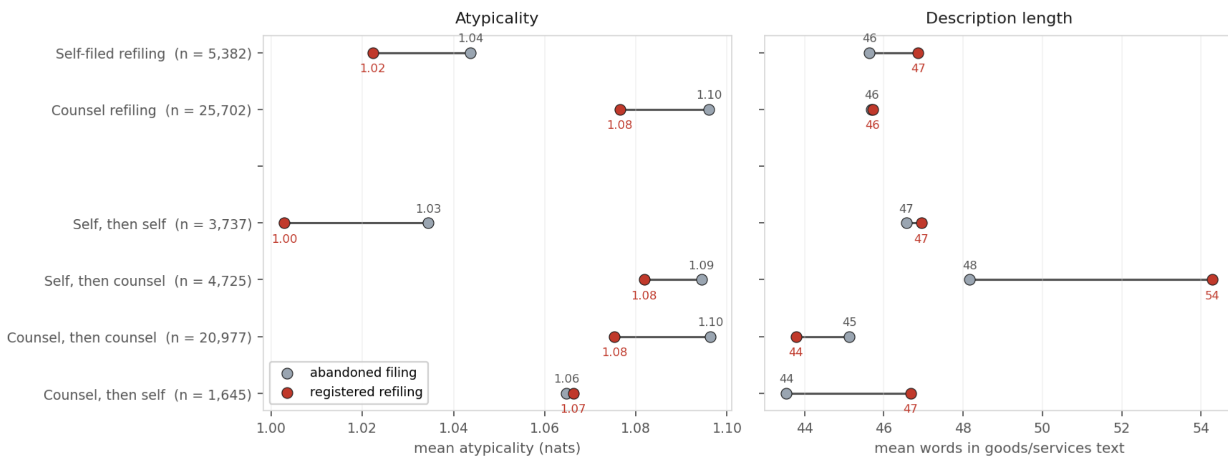

Figure 20: Rewritten refilings: mean atypicality and mean description length of the abandoned filing (grey) and of the registered refiling (red), by representation on the refiling and by the change in representation between the two attempts. Mean atypicality falls by 0.01 to 0.03 nats in every group except counsel-then-self, where it is unchanged. Length rises by six words where counsel is hired for the second attempt and falls by one where counsel rewrites counsel.

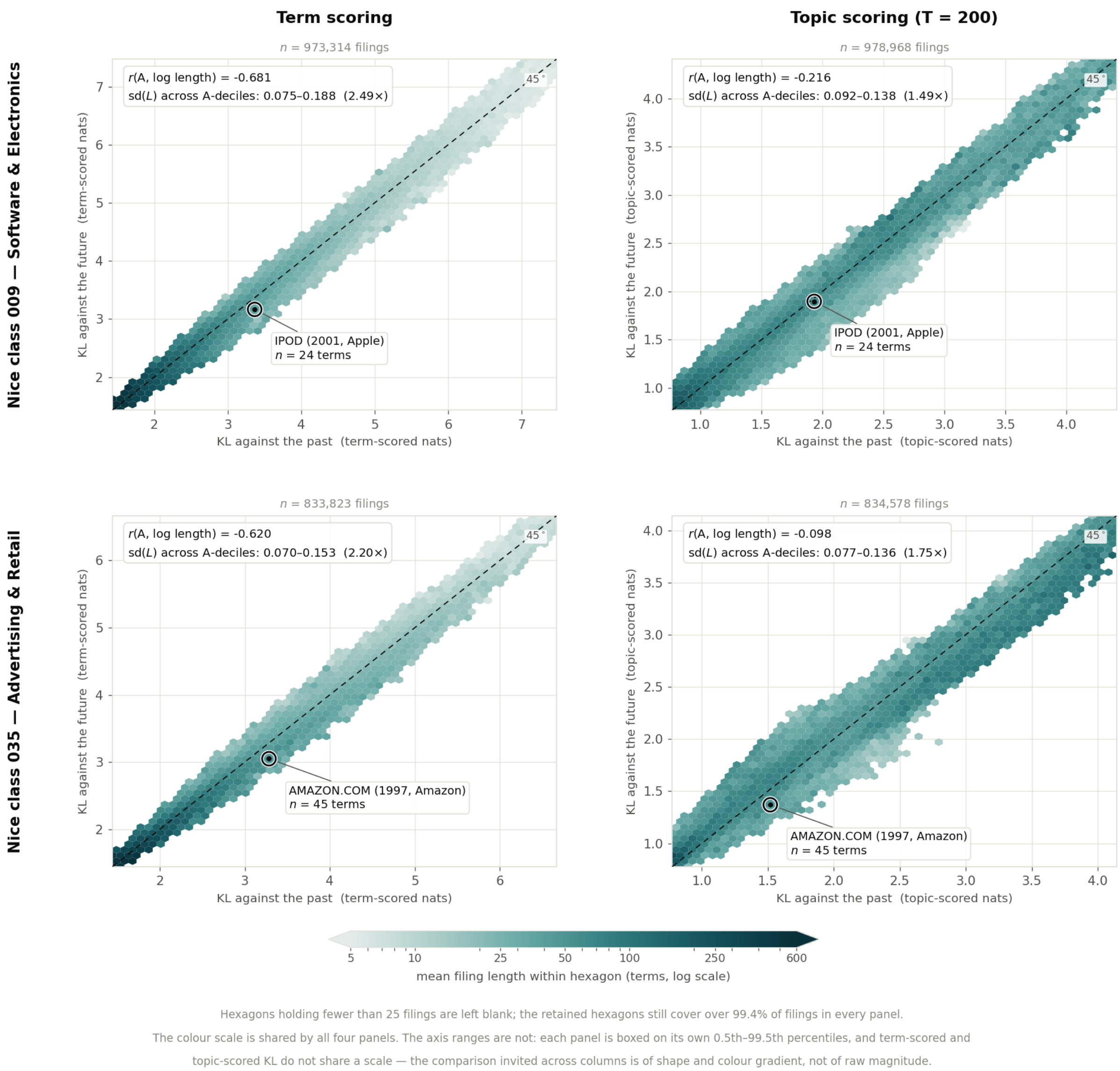

Figure 21: Mean filing length across the two axes, by representation and class, for filings 1995–2019 scored on per-filing reference windows. The horizontal axis is surprise against the preceding five years, the vertical axis surprise against the following five; a filing on the dashed diagonal is equally far from both, and distance below it is lead. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, since term-scored and theme-scored KL do not share a scale, axis ranges differ across columns because term- and theme-scored KL are not on a common scale, and only shape and colour gradient are comparable. Both columns are drawn on the filings the per-filing windows score, which differ between representations by under one percent.